

Computing & Robotics

Year 1

Student Manual



Computing & Robotics

Year 1 · Student Book

A robot does exactly what you say. Not what you meant.

INSIDE THIS BOOK

- Every Year 1 computing topic, in five units
- Predict, run and compare on every unit
- Real bug hunts, with the fix never printed
- Projects along the way, and one big one at the end
- QR codes, each printed in words as well

IMPRINT

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ABOUT THIS BOOK • CONTINUED

The block colours printed in this book match the real Scratch editor, so that a child who has met a block here will recognise it on screen.

Every character, story, fact card and illustration in this book was written and drawn for Prime School.

For teachers

This book is meant to be written in. Pupils will draw arrows, tick boxes, circle pictures and colour in these pages, and that is the point.

There are **no printed answers** anywhere in this book. Let students say what they think out loud first, then choose, circle or write. Read every instruction aloud together before pupils begin.

Roughly half the tasks in this book are marked **No screen needed**. Those are not a fallback for when the tablets are charging: unplugged work is where five- and six-year-olds actually learn to sequence, spot an error and fix it.



Chip prints every fact card in this book. Look for the torn paper tape.

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What is inside?



Five places in the yard. You will work in all five of them this year.

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WELCOME

There is a robot in the yard

Behind the school there is a **yard**. In the yard there is a bench, a shed, a table, a garden and a machine room. And in the yard there is **Beep**.

Beep is a robot. Beep is friendly, Beep is fast, and Beep will do **anything** you ask. That sounds wonderful. It is also the problem.



The one big idea in this book

A robot does exactly what you say. Not what you meant.

Say **walk** and Beep walks. Say walk again, and Beep walks again, straight into the wall, because you never said stop. Beep is not being silly. Beep is being **exact**.

So this year you will learn to say **exactly** what you mean. That is what computing is.

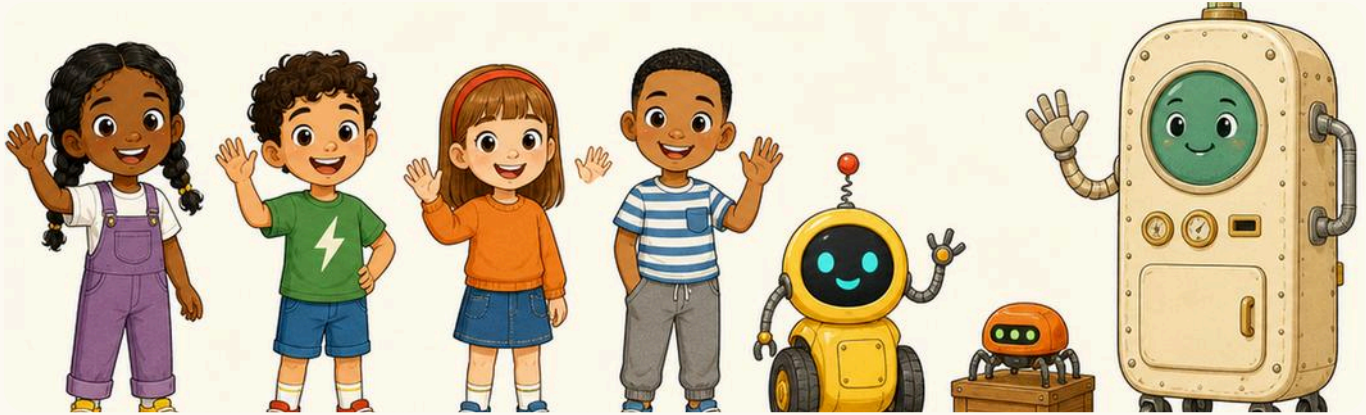
🚩 Chip's fact card

A person can hear **shut the door** and know to use the door they came in by, not the cupboard door. A computer cannot guess like that. Everything a computer does had to be said to it first, in order, by somebody.



Beep did exactly what it was told, and stopped when it reached the box. Not before.

Meet the team



Ada, Tomás, Inês, Rui, Beep, Bolt and Chip. You will meet all seven of them in every unit.

Ada

Ada writes every step down in her notebook before she starts.

Tomás

Tomás tries it first and fixes it after. Both ways work.

Inês

Inês likes things in order, and she notices when one is missing.

Rui

Rui always asks **what if**. He also loses his socks.

Beep, Bolt and Chip

Beep is the yard robot. Beep does exactly what you say. **Bolt** is small, has six legs, and carries things. **Chip** is the old thinking machine in the corner. When something surprising is true, Chip prints it on a little card for you.

Signs to look for

The same signs come back on every page. Once you know them, you always know what to do next.



Switch on

Something to talk about before you begin.



A quick game with your body and your voice.



Here is how it works

The teaching part. Read it out loud together.



Chip's fact card

Something true and surprising, printed by Chip.

HOW TO USE THIS BOOK • CONTINUED



Find the bug

One step here is wrong. Your job is to catch it.



Robot challenge

A harder task. Take your time.

1

Your turn means it is your turn to do something. A dashed box is always yours to write or draw in. When a task has a green edge like this one, it needs **no screen at all**.

No answers are printed in this book. You say what you think first, then choose.

NO SCREEN NEEDED

Come back to this page any time you forget what a sign means.

Five places to work



The yard from above. One place for each unit, in the order you will use them.

1 The Instruction Shed • say it in the right order

2 The Block Bench • build it out of blocks

3 The Sorting Table • find out what is true

4 The Wire Garden • join up and send a message

5 The Machine Room • look inside the machine

1 Point at each place on the picture as your teacher reads its name. Which one do you most want to try?

NO SCREEN NEEDED

Before you start



Your yard kit. You will not need all of it at once.

✓ Four rules of the yard

1. **Say it, then do it.** Plan out loud before you touch anything.
2. **Dry hands, clean table.** Water and crumbs are a machine's enemies.
3. **Carry a tablet flat, with two hands.** Never by one corner.
4. **A mistake is information.** When something goes wrong you have just found out something true. Say **good, now I know.**

The two buttons you will use most



green flag: GO



red button: STOP

Green means start. Red means stop. They look the same in this book as they do on the screen, so you will know them straight away.

For teachers

Before beginning Unit 1, check that pupils can follow a three-step spoken instruction in order, and can point to left and right on their own body. Both matter far more to early programming than any vocabulary.

Unit 1 needs no devices at all. If a floor robot is available it is ideal for Unit 1, but a chalked grid and a child pretending to be Beep works just as well, and pupils often understand it better.

1

UNIT 1

The Instruction Shed

Computational thinking and programming

In the shed you learn to say things in the right order, and what to do when one of them is wrong. It is the most useful thing in the whole yard.

1 Learn what an algorithm is

2 Use an algorithm

3 Learn what an error is

4 Find an error in an algorithm

What is an algorithm?



Before we begin

Tell your partner how you clean your teeth. Say every single thing you do. Did you remember to put the toothpaste on the brush? Did you remember to open the tube first?



By the end you will

- ✓ say what an algorithm is
- ✓ put steps in the right order
- ✓ spot a step that is missing
- ✓ make an algorithm of your own



Here is how it works

An algorithm is a list of steps, in the right order, that gets a job done.

You already know hundreds of them. Getting dressed is an algorithm. Making a sandwich is an algorithm. So is tidying the yard.

warm
up

Stand up. Now do exactly three things in this order: clap, turn round, sit down. Do them again in a different order. Was it still the same job?

Four steps make a sandwich



An algorithm for a jam sandwich. Four steps, and they only work in this order.

- 1 Take two slices of bread.
- 2 Spread the jam on one slice.
- 3 Put the other slice on top.
- 4 Eat it.

1.1 • HERE IS HOW IT WORKS • CONTINUED

- 1 Read the four steps out loud with your teacher. Now point at the one that must come **first**.

NO SCREEN NEEDED

- 2 What would happen if you did step 4 before step 2? Tell your partner. Then laugh about it.

You would eat two slices of dry bread.



Chip's fact card

The word **algorithm** comes from the name of a mathematician, al-Khwarizmi, who lived about 1200 years ago and wrote books explaining how to work things out step by step. His name travelled all the way into your computer.

Put them in order



Here are four steps for planting a seed. They are in the **wrong** order. Write 1, 2, 3 and 4 in the boxes to put them right.

☐

Water it.

☐

Fill the pot with soil.

☐

Cover the seed with soil.

☐

Push the seed into the soil.

1.1 • YOUR TURN • CONTINUED

- 1 Say your order out loud to your partner before you write anything down. If you disagree, act it out and see who is right.
NO SCREEN NEEDED

- 2 Now draw the very first step in the box.

draw step 1

→ One step more

Think of something you do every morning. Can you say it in exactly four steps? Not three. Not five. Four.

The words we use in the shed



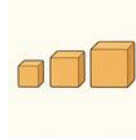
algorithm

steps in order



step

one thing you do



order

which comes first



plan

think before you start

- 1 Say each word out loud three times. Once quietly, once normally, and once like a robot.

NO SCREEN NEEDED

- 2 Copy these two words. Say each one before you write it.

1.1 • YOUR TURN • CONTINUED



step



plan

An algorithm is not a machine and it is not a computer. It is just a good list.

⚙️ MORE WORDS FROM THIS PAGE



first



next



last

Now give the steps to Beep



In the yard today

Beep is standing on the mat, waiting.

Beep cannot see where you want to go. Beep cannot guess. The only thing Beep can do is follow the cards you give it, one at a time, in the order you put them down.



Four cards is all Beep has

Beep understands exactly four instructions. That is all. Everything Beep ever does is made out of these.

1.2 • SWITCH ON • CONTINUED



forward



back



turn left



turn right

1

Hold up your left hand. Now your right. Beep needs you to be sure about this, so practise until you are.

NO SCREEN NEEDED



Chip's fact card

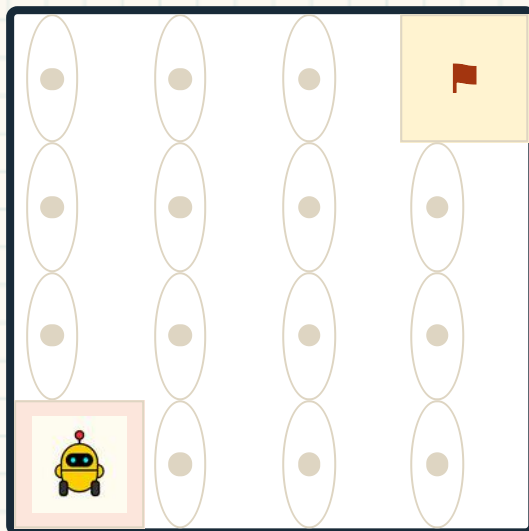
A real floor robot in a classroom usually knows only five or six instructions, just like Beep. Everything a big computer does is also built from a small handful of very simple steps. There are just an enormous number of them.

1.2 • HERE IS HOW IT WORKS

Six moves to the flag



Beep starts on the blue square. The flag is on the yellow square. Follow the arrows with your finger and see where Beep ends up.



1.2 • HERE IS HOW IT WORKS • CONTINUED

Beep starts here, bottom left. The flag is top right.

The algorithm



- 1 Put your finger on Beep. Now move it one square for each arrow, saying the arrow out loud. Did you reach the flag?

NO SCREEN NEEDED

→ One step more

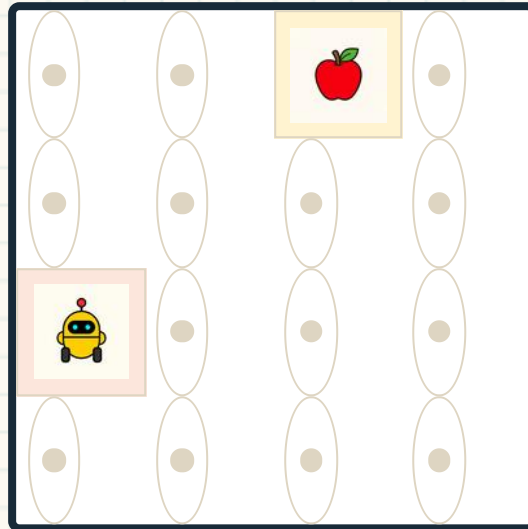
There is more than one way to get there. Could you go right first and up afterwards? Try it with your finger. How many moves is it that way?

1.2 • YOUR TURN

Now you write the algorithm

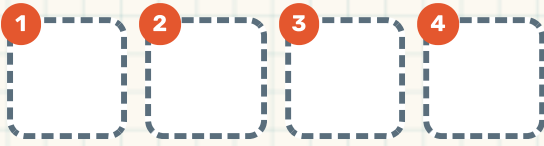


Beep is on the blue square again. The apple is what Beep is going to fetch. Draw one arrow in each empty card.



1.2 • YOUR TURN • CONTINUED

Four moves will do it. Count the squares first.



- 1 Count how many squares Beep must go **up**. Then count how many squares Beep must go **across**. Now draw them.

NO SCREEN NEEDED

- 2 Swap books with your partner and follow their arrows with your finger. Do they work?

Robot challenge

Give Beep a route that reaches the apple but uses **more** than four moves. A longer way is still a right answer, as long as Beep gets there.

What is an error?

In the yard today

Tomás said **forward** four times. Beep only needed three.

So Beep went forward four times, because that is what Tomás said. And on the fourth one, Beep rolled gently into a cardboard box and stopped.



Beep did nothing wrong. Beep did exactly what it was told.

Here is how it works

An error is a step that does not do what you wanted. The robot is never wrong. The **instruction** was wrong. That is good news, because you can change an instruction.

Chip's fact card

Programmers call a mistake in an algorithm a **bug**. In 1947 a team working on a huge early computer found a real moth stuck inside it, taped it into their notebook, and wrote that they had been **debugging** the machine. The name stuck.

Say exactly what you mean



Rui said **put on your socks**. He did not say **on your feet**.

- 1 Look at the picture. What did Rui **mean**? What did he **say**?
They are not the same.

NO SCREEN NEEDED



Inês said **pour the milk**. She forgot to say where.

- 2 Fix Inês's instruction. Say it out loud so that even Beep could not get it wrong.

Pour the milk into the cup.

Beep is not naughty. Beep is exact.

⚙️ WORDS FOR WHEN IT GOES WRONG



error



bug



fix

1.3 • YOUR TURN

Is this instruction exact enough?



Read each one. Circle **YES** if Beep could follow it without guessing.
Circle **NO** if Beep would need to guess.

Go forward three squares.	YES / NO
Go over there.	YES / NO
Turn right.	YES / NO
Turn a bit.	YES / NO
Stop.	YES / NO
Be careful.	YES / NO

- 1 For every **NO**, say a better instruction out loud. Make it so exact that Beep could not possibly get it wrong.

NO SCREEN NEEDED

→ **One step more**

Be careful is a lovely thing to say to a person and a useless thing to say to a robot. Why? Talk about it.

Finding the bug



Ada found the wrong card. She did not start again: she swapped one card.



Three steps to find a bug

1. **Run it slowly.** Do one step at a time and watch.
2. **Find the first step that goes wrong.** Not the last one. The first.
3. **Change that one step.** Then run it again.



Find the bug

Here is an algorithm for cleaning your teeth. One step is in the wrong place. The ladybird is sitting on it.

1 Pick up your toothbrush.

2 Put the toothbrush in your mouth.



3 Put toothpaste on the brush.

4 Brush all your teeth.

1 Which two steps need to swap? Say the numbers out loud.

NO SCREEN NEEDED

Now you find it

This algorithm is for putting on your shoes. This time there is **no ladybird**. One step is wrong, and you must find it yourself.

- 1 Sit down on the bench.
- 2 Put your shoes on your feet.
- 3 Put your socks on your feet.
- 4 Do up the straps.

- 1 Draw a ring round the step that is wrong. Then write the number it **should** be on the line under it.

NO SCREEN NEEDED

- 2 Act it out with your partner in the wrong order. It should feel silly. Silly is how you know you have found the bug.

Robot challenge

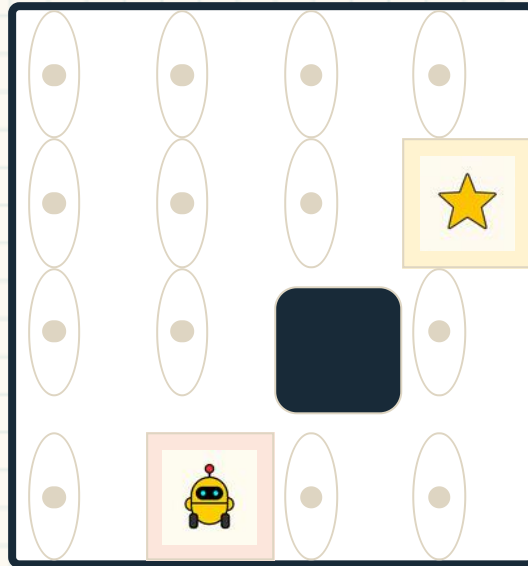
Write your own algorithm with a bug in it, in four steps. Give it to your partner. Can they find your bug?

1.4 • YOUR TURN

A bug on the mat



Beep should reach the star. But Beep ends up in the wrong place.
Follow the arrows with your finger and find out where it goes wrong.



The black square is a wall. Beep cannot go through it.



1.4 • YOUR TURN • CONTINUED

- 1 Move your finger one square for each arrow. Which arrow takes Beep into the wall?

NO SCREEN NEEDED

- 2 Write a better algorithm in these cards. Beep must not touch the wall.



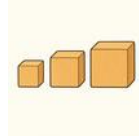
When a route hits a wall, do not throw the whole plan away. Change one arrow.

For teachers

This page is the single most important idea in Unit 1: **debugging is editing, not restarting**. Pupils will want to sweep all the cards away and begin again. Ask them each time: which is the FIRST card that is wrong? Then change only that one.

The words from Unit 1

Point at each picture and say the word. These are the words you will use every time you plan something.

**algorithm****step****order****instruction****robot****grid****error****bug****debug****fix****plan****sequence****1**

Choose two words. Say a whole sentence with each one in it.

NO SCREEN NEEDED

Make a route for Beep



Your class is going to build one long route across the yard and programme Beep to drive it, using nothing but cards.

- 1 Chalk a grid of squares on the ground outside, or lay out floor tiles. Four squares by four is plenty.
- 2 Put Beep on one corner and something to fetch on another. Agree where Beep must end up.
- 3 Lay your instruction cards on the ground in a line, in order. Say each one out loud as you put it down.
- 4 Run it slowly, one card at a time. When Beep goes wrong, find the FIRST wrong card and change only that one.

→ One step more

No floor robot? One person is Beep. They must do exactly what the cards say and nothing else. It is even funnier.

Think about this unit

Colour one circle for each line. Green means yes, yellow means nearly, red means not yet. Not yet is a perfectly good answer: it tells you what to practise next.

I can say what an algorithm is.	☐ ☐ ☐
I can put four steps into the right order.	☐ ☐ ☐
I can write a route for Beep on a grid.	☐ ☐ ☐
I can find the step that is wrong.	☐ ☐ ☐
I can fix one step instead of starting again.	☐ ☐ ☐

Now I can

- ☐ say what an algorithm is
- ☐ put steps in the right order
- ☐ give Beep a route with arrow cards
- ☐ find an error and fix it

Next, at the Block Bench, you will build your algorithms out of blocks.

2

UNIT 2

The Block Bench

Coding and programming

At the bench you build your algorithms out of coloured blocks that snap together. Each colour makes a different kind of thing happen.

1 Scripts, and how blocks snap together

2 Motion, Looks and Sound

3 Events: making things start

4 Control: repeat, and if

Blocks that snap together



Before we begin

Have you ever built something out of bricks that click together? What happens if you try to click two bricks together the wrong way up?



By the end you will

- ✓ snap blocks together into a script
- ✓ read a script from the top downwards
- ✓ name the colour of each kind of block
- ✓ make things move, change and make a sound

2.1 • SWITCH ON • CONTINUED



Every block has a notch on top and a bump underneath, so they only join one way: downwards.



Here is how it works

A **block** is one instruction. When you snap blocks together you get a **script**. A script runs from the **top** to the **bottom**, one block at a time, exactly like your cards in the shed.

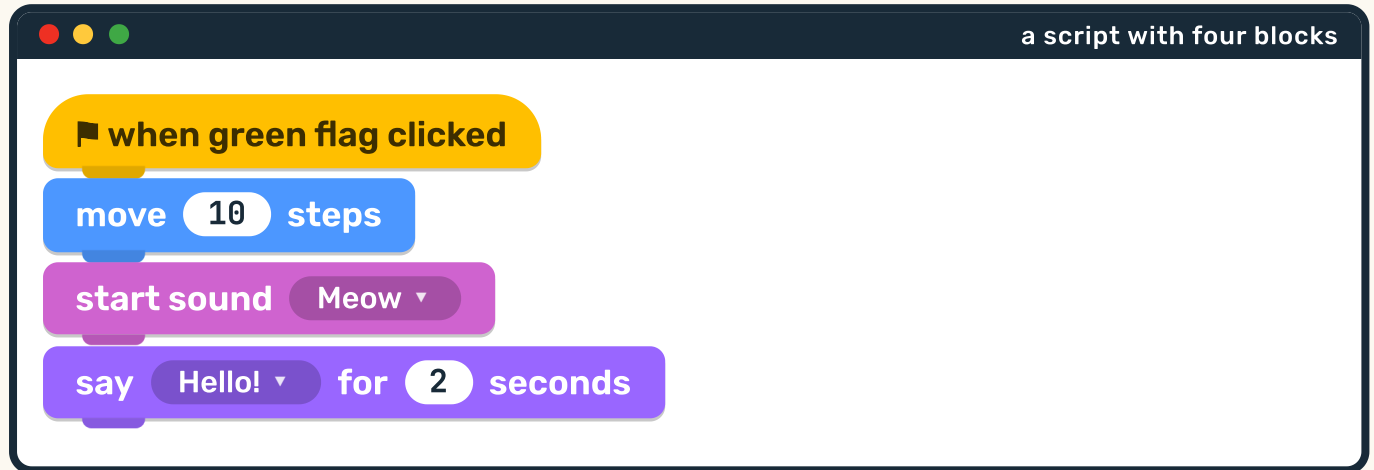


Chip's fact card

The colours are not decoration. Every colour is a different **drawer** in the toolbox. Once you know that blue means moving, you can find the block you want without reading a single word.

A script is a stack

Here is a real script. Read it from the top. Put your finger on each block and say what it does.



2.1 • HERE IS HOW IT WORKS • CONTINUED

1 How many blocks are in this script? Point and count. NO SCREEN NEEDED

2 Which block happens **first**? Which happens **last**? How do you know?

Top block first, bottom block last. Always.

→ One step more

Look at the four colours in the script above. Four colours means four different drawers. You will open every one of them in this unit.

2.1 • YOUR TURN

The nine drawers

These are the nine kinds of block you will meet. The colours here are exactly the colours you will see on the screen.



- 1 Say the colour of the drawer you would open to make something **move**. Now to make a **sound**.

NO SCREEN NEEDED

- 2 Colour these three boxes: one blue for Motion, one purple for Looks, one pink for Sound.



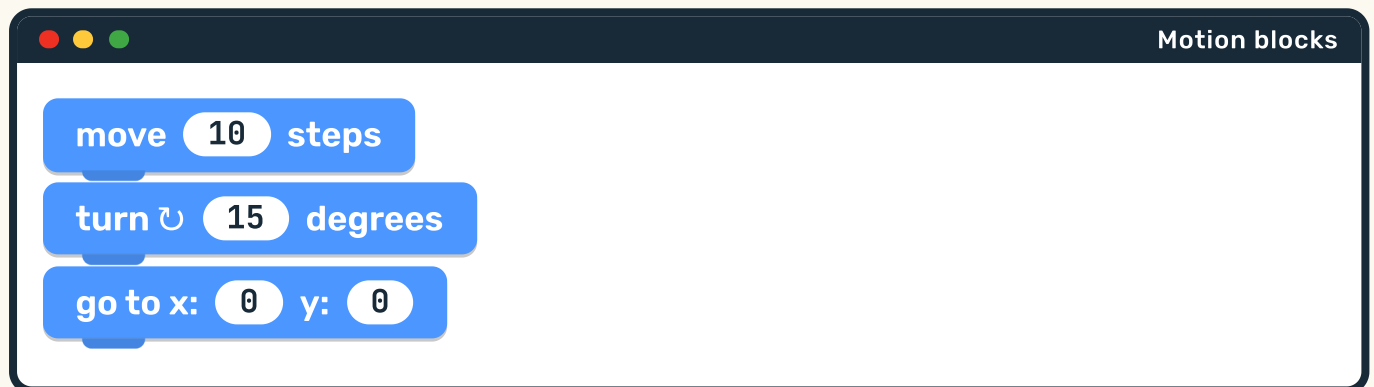
Making the cat move

In the yard today

The cat on the screen is called a **sprite**.

A sprite is anything on the screen that you can tell what to do. It might be a cat, a ball, a rocket or a robot. The blue blocks are the ones that make a sprite move.

These are **Motion** blocks. Blue means moving.





Here is how it works

move 10 steps makes the sprite slide forward a little way. Change the 10 to a bigger number and it goes further. Change it to a smaller number and it creeps.

1

Stand up. Take 10 tiny steps. Now take 10 giant steps. Same number, very different journey. A computer is the same.

NO SCREEN NEEDED



Chip's fact card

A **step** on the screen is tiny: about the width of one dot of light. That is why **move 10 steps** looks like a small hop rather than a walk across the room.

Bigger number, longer jump



which one goes further?

move 50 steps

move 5 steps



- 1 Circle the block that makes the cat go **further**.

NO SCREEN NEEDED

- 2 Write a number in this block to make the cat move a really long way.

2.2 • YOUR TURN • CONTINUED

move steps



Find the bug

Ada wanted her cat to go all the way across the screen. She used **move 1 steps** and pressed go. The cat did move. She just could not see it.

3

What should Ada change? Say the new number out loud.

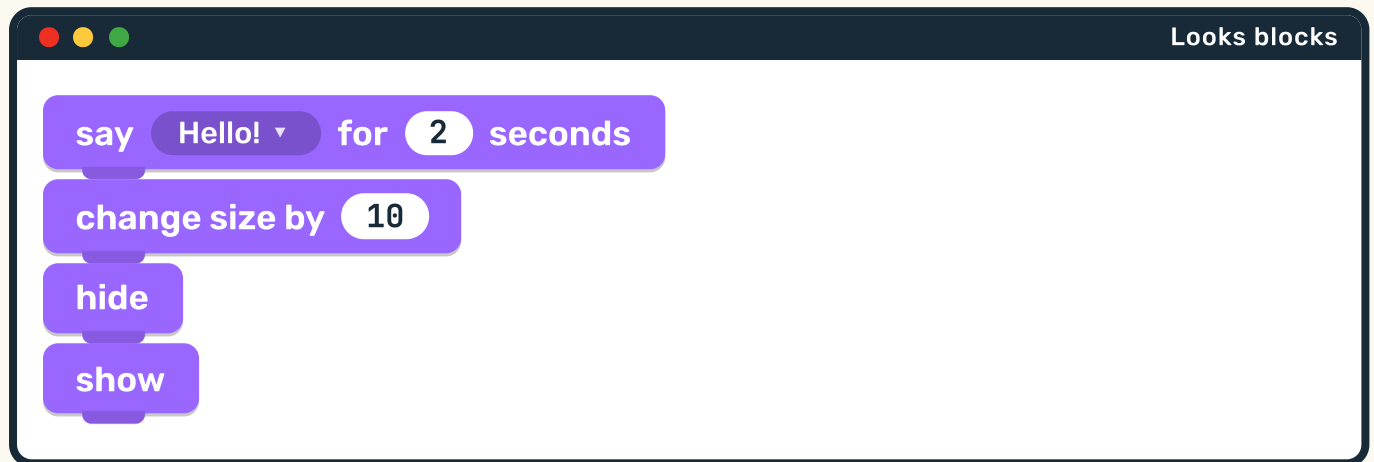
NO SCREEN NEEDED

→ One step more

Turning is measured in **degrees**. A quarter turn is 90 degrees. Stand up and make a quarter turn. Now do four of them. You are facing the front again.

Changing how it looks

These are **Looks** blocks. Purple means how something looks: its size, its colour, whether you can see it at all, and what it says.



the cat as it starts



after **change size by 50**

2.3 • LOOKS • CONTINUED

- 1 Which block would you use to make the cat **bigger**? Point at it.
NO SCREEN NEEDED

hide does not delete your sprite. It is still there, just invisible. **show** brings it straight back.

Chip's fact card

A **say** block puts a real speech bubble on the screen, just like the ones in a comic. Whatever you type inside the block is what appears in the bubble.

What will the cat say?



Write your own words in the speech bubble. Then say them out loud in a cat voice.



After **hide**, the cat fades away. It has not gone.

write what the cat says

- 1 Now write the same words inside this block, so the computer knows them too.

NO SCREEN NEEDED

2.3 • YOUR TURN • CONTINUED

say for 2 seconds

- 2 Sort these three jobs into the right drawer. Write M for Motion or L for Looks beside each one.

NO SCREEN NEEDED

☐

make the cat slide forward

☐

make the cat twice as big

☐

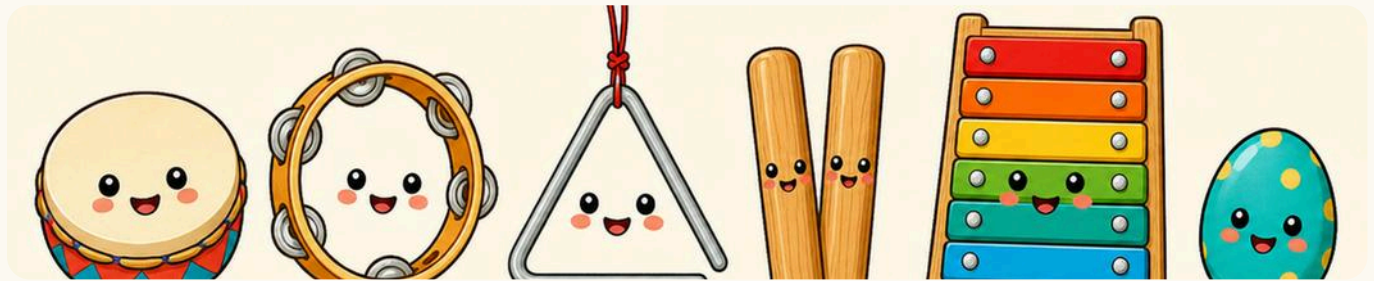
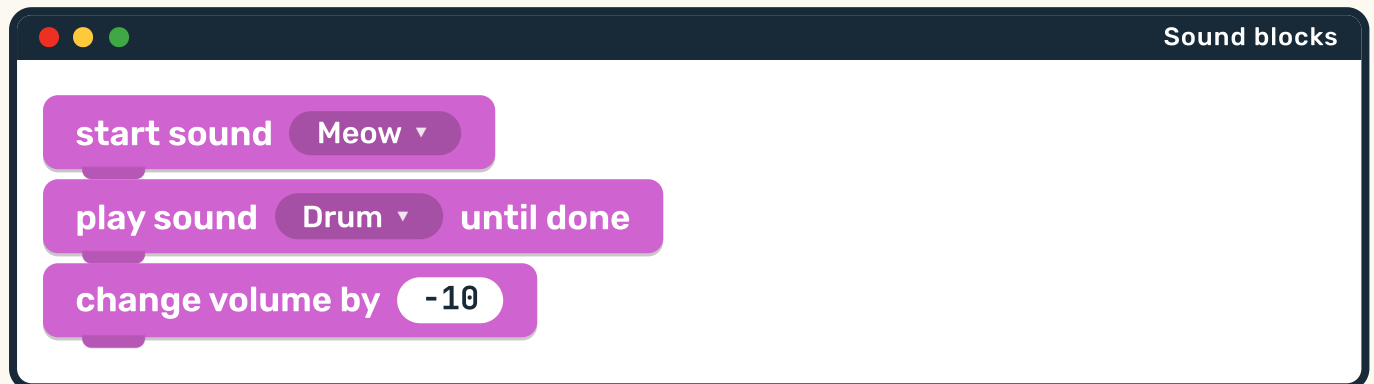
make the cat disappear

Robot challenge

Make a script with a Looks block AND a Motion block in it. Draw them in the right order, and say what will happen.

Making a noise on purpose

These are **Sound** blocks. Pink means noise: playing a sound, and making it louder or quieter.



Six sounds you can make in the classroom without any screen at all.



Two sound blocks, two different jobs

start sound begins the noise and moves straight on to the next block. **play sound until done** waits for the noise to finish first. One is a shout while you keep running. The other is a shout, then run.

1

Try it with your body. Clap and keep walking. Now clap, wait for the sound to stop, then walk. Which felt slower?

NO SCREEN NEEDED

Loud, quiet, loud again



Volume is how loud something is. It has a number, from 0 to 100.

- 1 Say your own name at volume 100. Now at volume 10. Now at volume 0. What happened at 0?

Nothing at all. 0 is silence.

NO SCREEN NEEDED

- 2 Circle the picture of the sound you would choose for a robot walking.



drum



loud



quiet

⚙️ **SOUND WORDS**



sound



loud



quiet

→ **One step more**

A sound with volume 0 is still playing. The computer is still doing the work. You just cannot hear it. Is that the same as no sound at all? Talk about it.

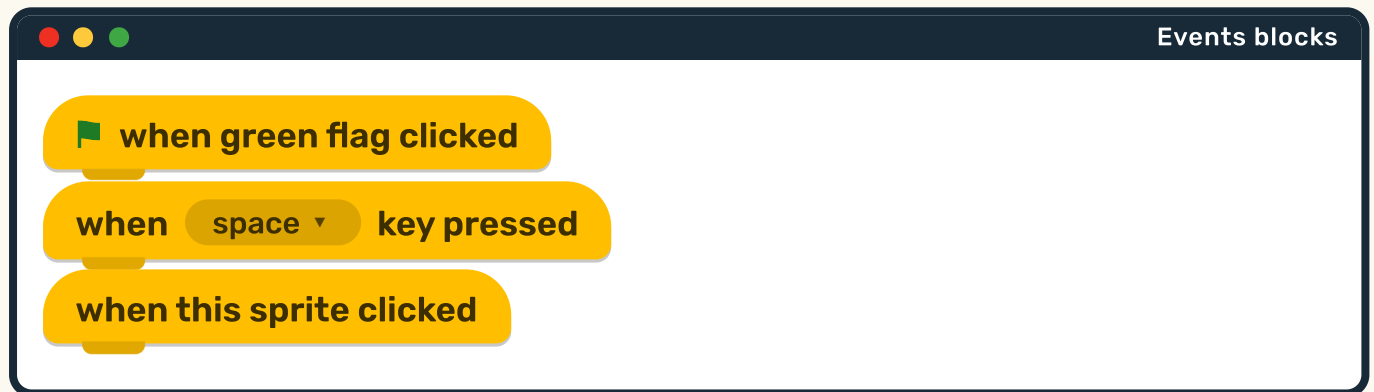
What makes it start?

In the yard today

A script sits there doing nothing until something wakes it up.

The thing that wakes it up is called an **event**. The most famous event of all is clicking the green flag.

These are **Events** blocks. Yellow means **when**. Look at the shape: a yellow block has a rounded top like a hat, so nothing can ever snap on above it. It must always go first.



2.5 • EVENTS • CONTINUED



Green means go. Red means stop. They are the same two buttons on every screen.

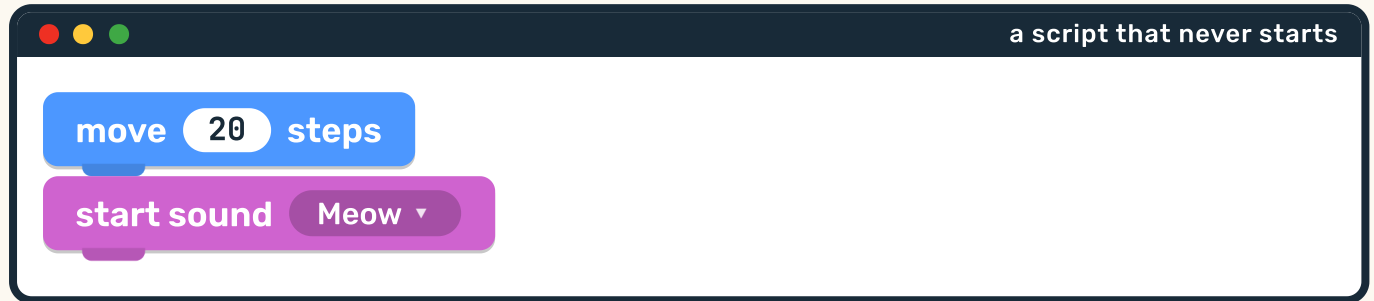
Chip's fact card

An Events block is called a **hat block** because it sits on top of the stack like a hat. Its top is round, so nothing fits above it. The shape itself tells you it goes first.

Give this script a hat



Here is a script with no hat block. Nothing will ever start it.



- 1 Which block is missing? Draw a yellow hat block on top of it. NO SCREEN NEEDED

draw the missing hat block

2.5 • YOUR TURN • CONTINUED



press this and the script runs



press this and everything stops

- 2 Play the flag game. When your teacher raises a green card, do your dance. When they raise a red card, freeze. You are a script now.

NO SCREEN NEEDED



Find the bug

Rui built a lovely script and pressed the green flag. Nothing happened at all. His hat block said **when space key pressed**.

- 3 What should Rui do? There are two right answers. Can you find both?

Press the space key, or change the hat block.

Doing it again, and again



Before we begin

Say **jump** four times out loud. Now think of a shorter way to ask somebody to jump four times, using fewer words.

These are **Control** blocks. Orange means **when** and **how many times**. A Control block is a different shape: it wraps round the blocks inside it, like a big C.



Round and round. A loop is a circle you come back to.

a repeat block

repeat 4

move 10 steps

turn 90 degrees



Here is how it works

Everything **inside** the C happens 4 times. So the sprite moves and turns, moves and turns, moves and turns, moves and turns. Then it stops, and whatever is below the C happens next.

1

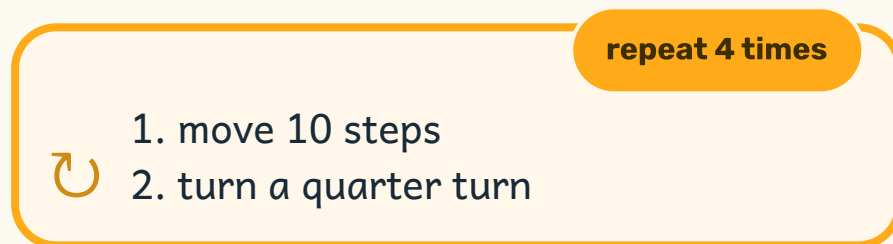
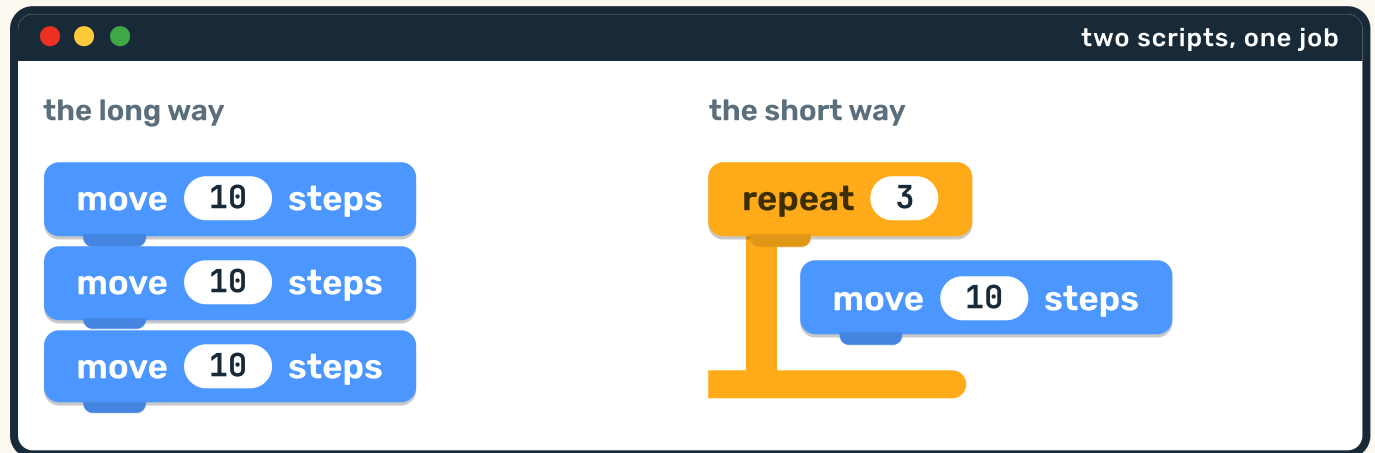
Move 10, turn a quarter turn, four times over. What shape does the sprite draw? Try it with your finger on the table.

A square.

NO SCREEN NEEDED

A loop is shorter than a list

These two do **exactly** the same thing. One of them is much shorter to write.



2.6 • HERE IS HOW IT WORKS • CONTINUED

- 1 Count the blocks in each script. Which one has fewer blocks?
NO SCREEN NEEDED

- 2 Clap 8 times. Now say the shortest instruction you can that means the same thing.
Repeat clap, 8 times.

Chip's fact card

Loops are why a computer can do a job a million times without getting bored or making a mistake. You write it once. It does it a million times, exactly the same.

If this, then that

Sometimes you only want something to happen **sometimes**. That is what an **if** block is for.

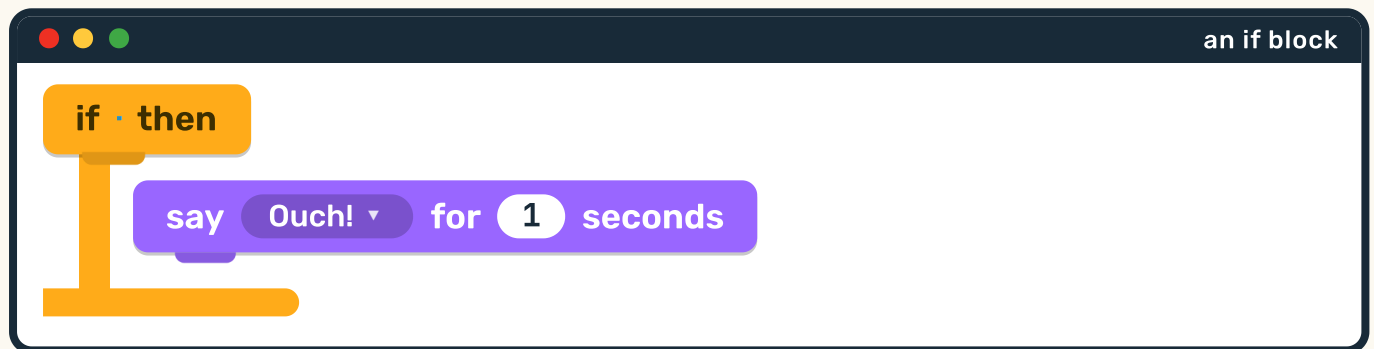
if **is it raining?** then

YES, DO THIS

Put your coat on.

NO, DO THIS

Leave your coat on the peg.



**Here is how it works**

An **if** block asks a question that can only be answered **yes** or **no**. If the answer is yes, the blocks inside happen. If the answer is no, they are skipped completely.

1

Play the if game. Your teacher says: **if you are wearing something red, stand up**. Only some of you will move. That is an if.

NO SCREEN NEEDED

2

Finish this one out loud: **if the bell rings, then ...**

→ One step more

What about **if it is raining, put your coat on, otherwise leave it on the peg**? That is an if with two paths. Programmers call the second path **else**.

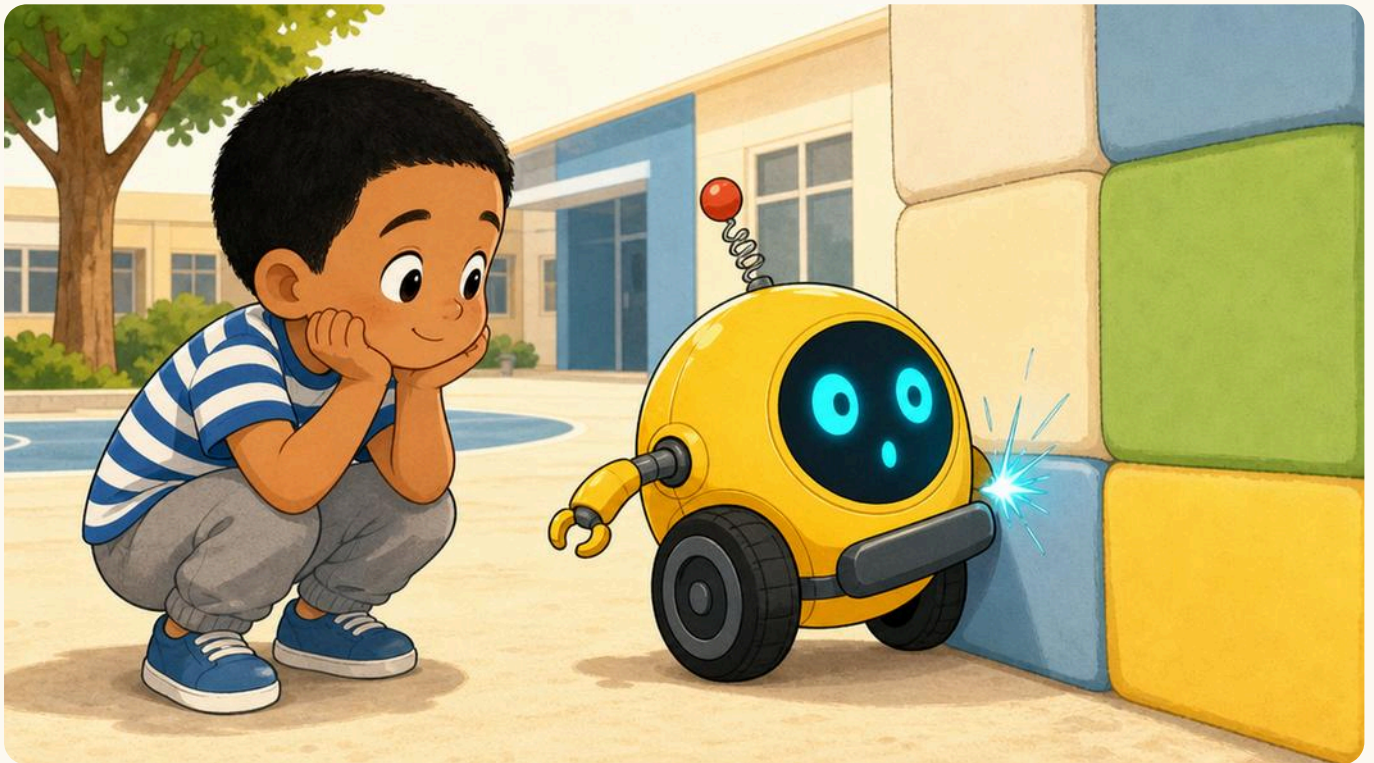
How does it know?

In the yard today

Beep rolled up to the wall and stopped all by itself.

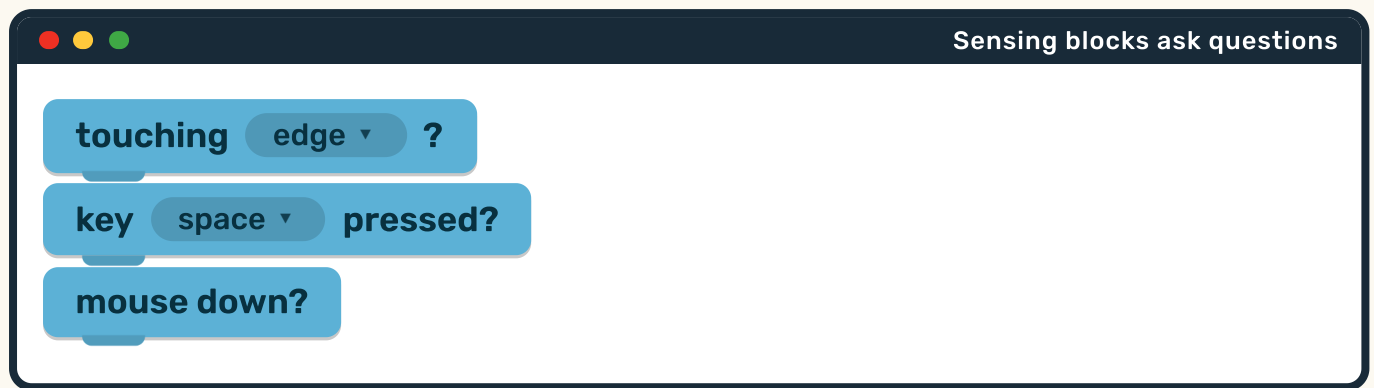
Nobody told Beep the wall was there. Beep **felt** it. A machine that can feel something has a **sensor**, and the blocks that use sensors are the light blue ones.

These are **Sensing** blocks. Light blue means **finding out**. Notice the shape: a sensing question is a stretched hexagon, because it is a question, not an instruction.



Beep's bumper is a sensor. When it is pressed, Beep knows.

2.7 • SENSING • CONTINUED



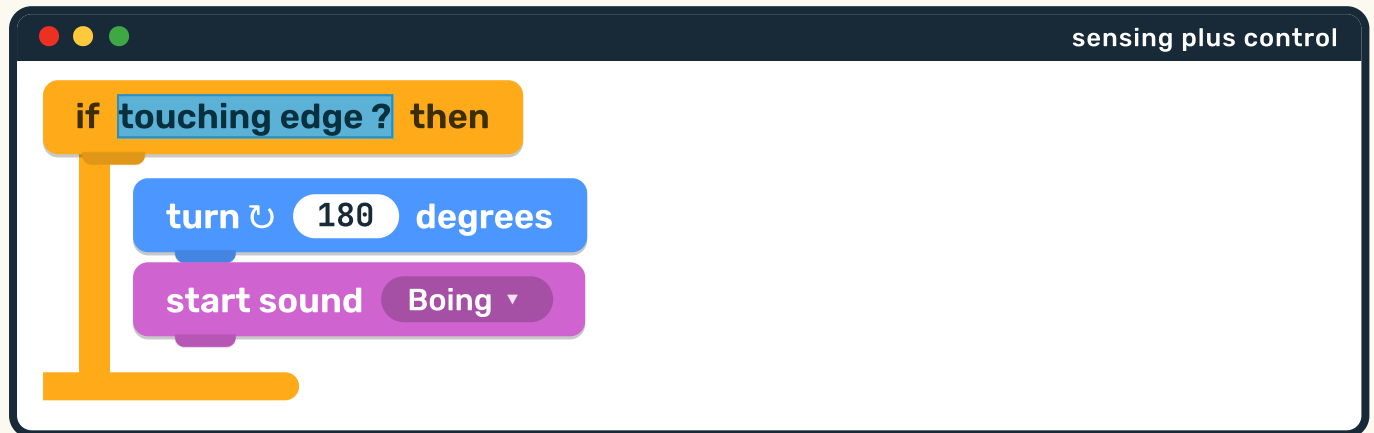
A sensor is the part that feels. This one has an eye and three lines showing what it can reach.

Chip's fact card

Your own body is covered in sensors. Shut your eyes and let a friend hold a hand near your cheek without touching it. Often you still know it is there, because your skin senses the warmth.

A question plugs into an if

On its own, a sensing question does nothing. It becomes useful when you drop it into the hole in an **if** block.



Read it out loud

If you are touching the edge, **then** turn all the way round and go boing. Otherwise, do nothing at all and carry on.

2.7 • HERE IS HOW IT WORKS • CONTINUED

- 1 Be the sprite. Walk slowly towards a wall with your hand out. The moment you touch it, turn right round. You have just run this script.

NO SCREEN NEEDED

- 2 Circle the sensing question in the script above. What shape is it?

A stretched hexagon.

→ One step more

What would happen if the **turn 180 degrees** block was missing?
Beep would touch the edge, go boing, and still be stuck against the edge. So it would go boing again. And again.

Which sensor would you use?



Draw a line from each job to the sensor that could do it. Say your answer out loud before you draw.

A light that comes on when somebody walks past.

A door that opens when you press a button.

A game that starts when you click the mouse.

a **movement** sensor

a **touch** sensor

a **mouse** sensor

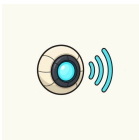
2.7 • YOUR TURN • CONTINUED

- 1 Walk round your school and find one real sensor. A tap that runs when you put your hands under it? A light that knows when it is dark? Draw what you found.

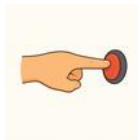
NO SCREEN NEEDED

draw the sensor you found

⚙ SENSING WORDS



sensor



touch



question

A box that holds one thing



Before we begin

How would you keep the score in a game of beanbags, without writing anything down? Could you use your fingers? What if the score got to twenty?

These are **Variables** blocks. Orange means a **variable**, which is a box with a name on it that holds **one** thing.



The peg board is a variable. It holds one number: the score.

Variables blocks

set score ▾ to 0

change score ▾ by 1

show variable score ▾



Set and change are not the same

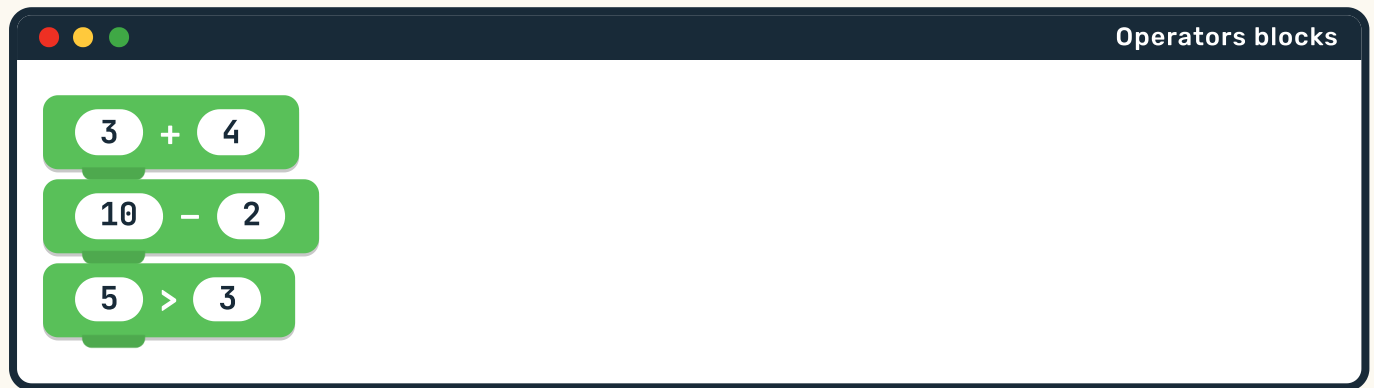
set score to 0 throws away whatever was in the box and puts 0 in.
change score by 1 keeps what is in the box and adds one more.
Use **set** at the start of a game. Use **change** every time somebody scores.

- 1 Hold up 3 fingers. Now **change by 1**. Now **change by 1** again.
Now **set to 0**. What happened to your fingers?

NO SCREEN NEEDED

Doing sums to what is in the box

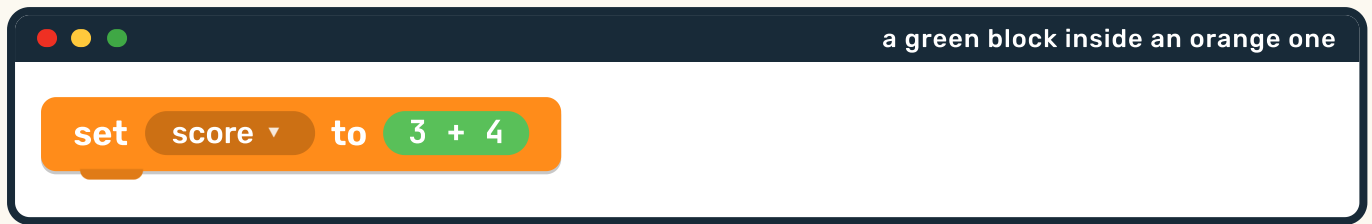
These are **Operators** blocks. Green means **working something out**: adding, taking away, and comparing two numbers to see which is bigger.



Here is how it works

A green block does not **do** anything on its own. It works out an answer and hands it over. So you drop a green block inside another block, where the number would go.

2.8 • OPERATORS • CONTINUED



- 1 What number goes into the score box? Say it out loud.

Seven.

NO SCREEN NEEDED

- 2 Work these out with your teacher. Write the answer in each box.

$$2 + 3 =$$

$$6 - 4 =$$

$$5 + 5 =$$

Chip's fact card

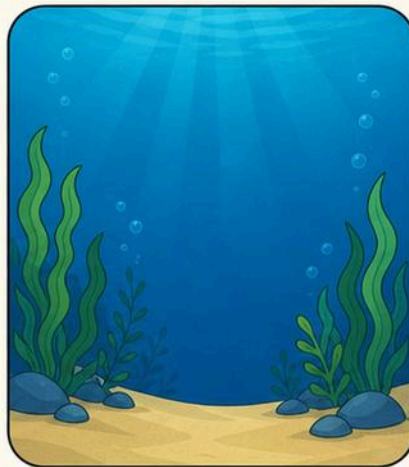
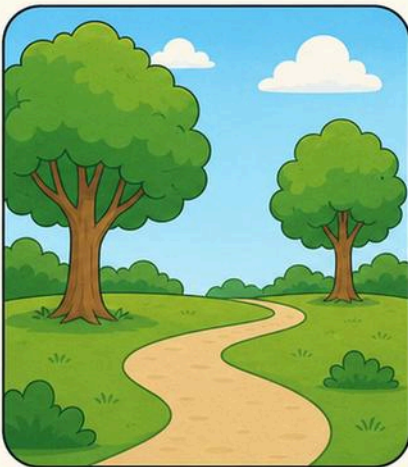
The **>** sign means **is bigger than**. Some people remember it as a hungry mouth: it always opens towards the bigger number.

Changing where the story happens

In the yard today

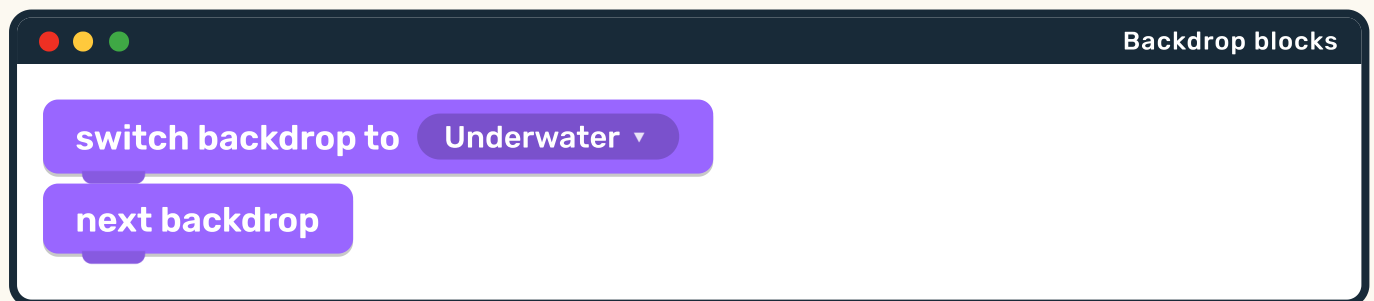
The cat is in a park. Now the cat is under the sea.

The cat did not move. The **backdrop** changed. A backdrop is the picture behind everything else, and swapping it changes the whole story.



Three backdrops: a park, under the sea, and a starry night.

Backdrop blocks live in the **Looks** drawer, because a backdrop is about how things **look**.



2.9 • BACKDROPS • CONTINUED

- 1 Look at the three backdrops. Which one would you choose for a story about a rocket? Which for a story about a fish?

NO SCREEN NEEDED

- 2 Hold up a sheet of paper behind a toy. Now swap it for a different colour. You have just switched the backdrop.

→ **One step more**

next backdrop means go to the one after this. If you are on the last one, it goes back round to the first. That is a loop hiding inside a Looks block.

Draw your own backdrop



A good backdrop has **nothing** important in the middle, because that is where your sprite is going to stand.

draw a backdrop for your story

- 1 Where will your sprite stand? Draw a small cross on the spot you left empty for it.

NO SCREEN NEEDED

**Find the bug**

Inês drew a beautiful backdrop of a forest, with an enormous tree right in the middle. Then she put her cat sprite on the stage. The cat was completely hidden behind the tree.

2

How could Inês fix this without drawing the whole backdrop again? Give two ideas.

Move the cat, or move the tree to one side.

NO SCREEN NEEDED

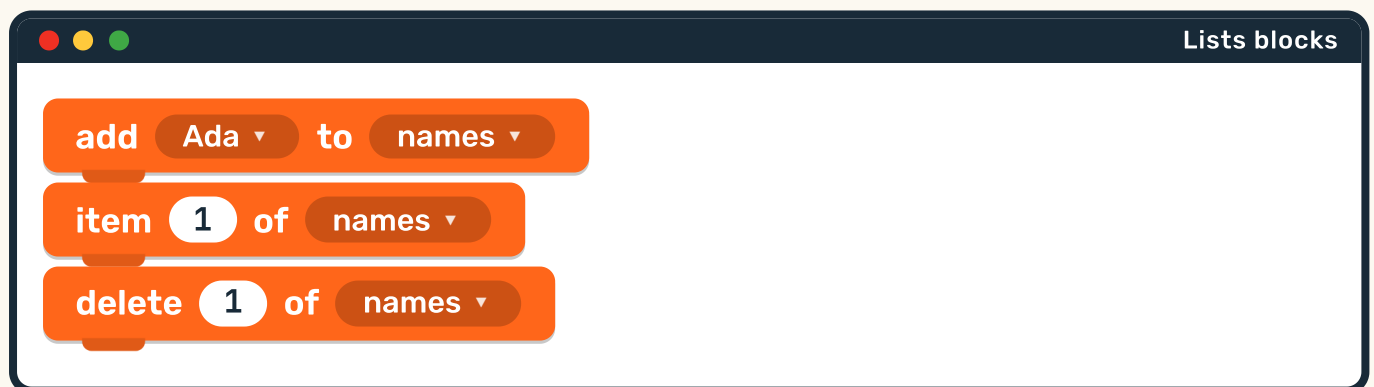
STAGE WORDS**backdrop****stage****sprite**

A box that holds many things

A variable holds **one** thing. A **list** holds **many** things, and it keeps them in order. These are **Lists** blocks: a darker orange, because a list is a variable's big brother.



Five cards on a string, in order. That is a list.



**The order matters**

Every thing in a list has a **number**: the first one is item 1, the next is item 2, and so on. So you can always ask for exactly the one you want.

a variable

score **7**

a list

names **1** Ada **2** Tomás **3** Rui

1

In the list above, who is item 2? Who would be item 4 if Inês joined?

NO SCREEN NEEDED

Make a list of your own

Write four things you do at school every day, in the order you do them. Number them as you go.

my list

1

Read your list to your partner. Now ask them: **what is item 3?**
They must answer without counting from the start.

NO SCREEN NEEDED

2.10 • YOUR TURN • CONTINUED

2 Cross out item 2 on your list. Now what number is the thing that used to be item 3?

It becomes item 2. Everything after a gap slides up.

NO SCREEN NEEDED

Robot challenge

Ten children are lining up. Is a line a list or a variable? How do you know? Say why out loud.

Chip's fact card

A shopping list on paper and a list inside a computer work the same way. Both keep things in order, and both let you add to the bottom or cross one out of the middle.

Which way will it go?



Before we begin

Think about walking to the school gate. Is there more than one way to get there? Does everybody in your class take the same one?



Here is how it works

Control flow means the path your script takes through its blocks. Sometimes it runs straight down. Sometimes it loops back up. Sometimes it splits in two and only one side happens.

2.11 • LOGIC AND CONTROL FLOW • CONTINUED



straight on

one block after another



round again

a loop



split in two

an if

if **is the score 10?** then

YES, DO THIS

Say **you win!** and stop.

NO, DO THIS

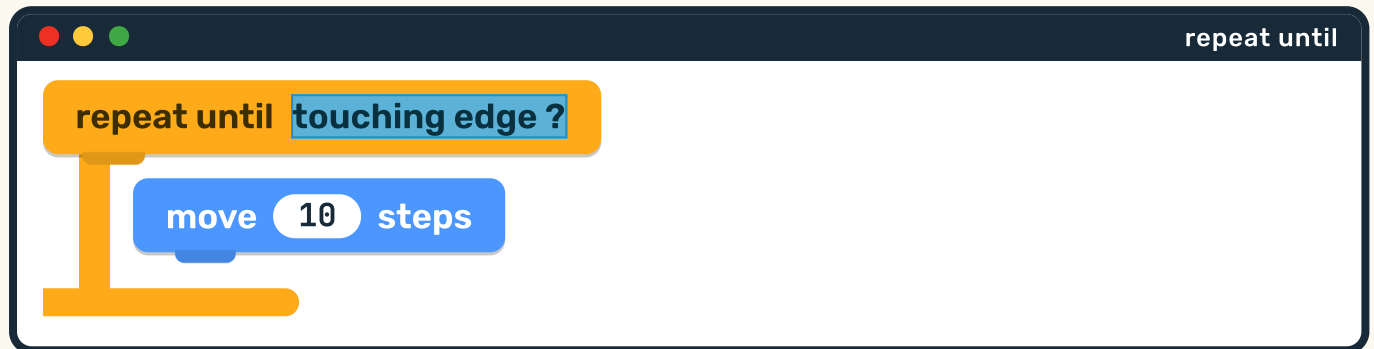
Carry on playing.

- 1 Trace the path with your finger: start at the top, reach the question, then choose one side. Which side did you take, and why?

NO SCREEN NEEDED

A loop with a question in it

This is the most useful shape in all of programming: keep going **until** something is true.



Read it out loud

Move 10 steps. Are you touching the edge? No? Then move 10 steps again. Are you touching the edge now? Keep asking, keep moving, and stop the moment the answer is yes.

- 1 Play it. Walk towards the wall in small steps. After each step, ask yourself out loud: **am I touching it?** Stop when the answer is yes.

NO SCREEN NEEDED



Find the bug

Tomás wrote **repeat until touching edge**, but he put **no** movement block inside it. His sprite was not touching the edge, so the loop asked the question again. And again. It never stopped.

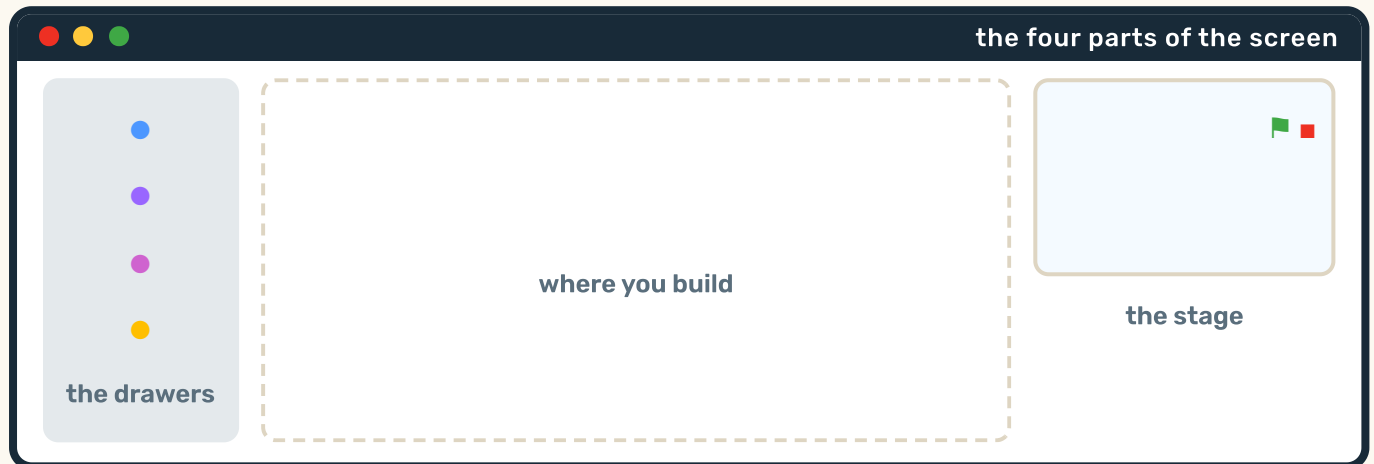
- 2 Why did Tomás's script get stuck forever? What must go inside a **repeat until** loop?

Something that changes, so the answer can become yes.

A loop that can never end is called an **infinite loop**. Every programmer in the world has written one by accident.

Finding your way round the screen

The screen is always laid out the same way. Once you know the four parts, you can find anything.



1. **The drawers** down the left. Coloured dots, one for each kind of block.
2. **The building space** in the middle. Drag blocks here and snap them together.
3. **The stage** at the top right. This is where your sprite lives and where you watch it work.
4. **The green flag and the red button**, just above the stage. Go, and stop.

1 Point at each of the four parts on the picture as your teacher names it. Then close the book and name all four from memory.

NO SCREEN NEEDED

Where would you click?

Read each job. Say which of the four parts of the screen you would go to. Then circle YES if you are sure.

You want a block that makes the cat move.	YES / NO
You want to watch the cat move.	YES / NO
You want to snap two blocks together.	YES / NO
You want everything to stop right now.	YES / NO

 go
  stop

- 1 Practise the two most important clicks in the whole book: green flag to go, red button to stop. Pretend to click each one now.

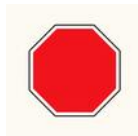
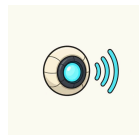
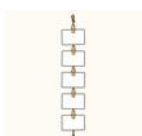
For teachers

Pupils reliably forget the red button exists, then panic when a script runs away. Make stopping a habit from the first lesson: every pupil presses stop before they leave the device, every single time.

If devices are shared, a two-minute rota with a timer works better than asking pupils to judge when their turn is over.

The words from Unit 2

Point at each picture and say the word. These are the words of the bench.

**block****script****sprite****stage****move****turn****costume****say****sound****event****green flag****stop****loop****forever****sensor****variable****backdrop****list**

- 1 Choose three words. Say one whole sentence that uses all three.

NO SCREEN NEEDED

Make the cat do a dance



You are going to build one script, from a hat block down to the last block, and make it run.

- 1 Plan it on paper first. Write down four things you want your sprite to do, in order. This is your algorithm.
- 2 Put a yellow hat block at the top: **when green flag clicked**. Nothing starts without it.
- 3 Snap your four blocks underneath, in your planned order. Use at least one Motion block and one Sound block.
- 4 Press the green flag. When it does not do what you meant, find the **FIRST** block that is wrong and change only that one.

→ One step more

Add a **repeat** block round your whole dance to make it go again and again. Then find out how to make it stop.

Think about this unit

Colour one circle for each line. Green means yes, yellow means nearly, red means not yet. Not yet is a perfectly good answer: it tells you what to practise next.

I can snap blocks together into a script.	☐ ☐ ☐
I know which colour drawer to open.	☐ ☐ ☐
I can put a hat block at the top.	☐ ☐ ☐
I can use a repeat block to make something happen again.	☐ ☐ ☐
I can find the four parts of the screen.	☐ ☐ ☐

Now I can

- ☐ build a script that runs from top to bottom
- ☐ name the nine kinds of block by colour
- ☐ make a sprite move, change and make a sound
- ☐ use repeat, and use if
- ☐ find my way round the screen

Next, at the Sorting Table, you will find out what is really true.

3

UNIT 3

The Sorting Table

Managing data

On the table you find out what is really true, instead of guessing. You collect it, you sort it, you draw it, and then you read what it says.

1 What is data

2 Collecting data

3 Sorting and grouping

4 Showing it, and reading it

What is data?



Before we begin

How many children in your class have brown eyes? Do not guess. How could you find out for certain?

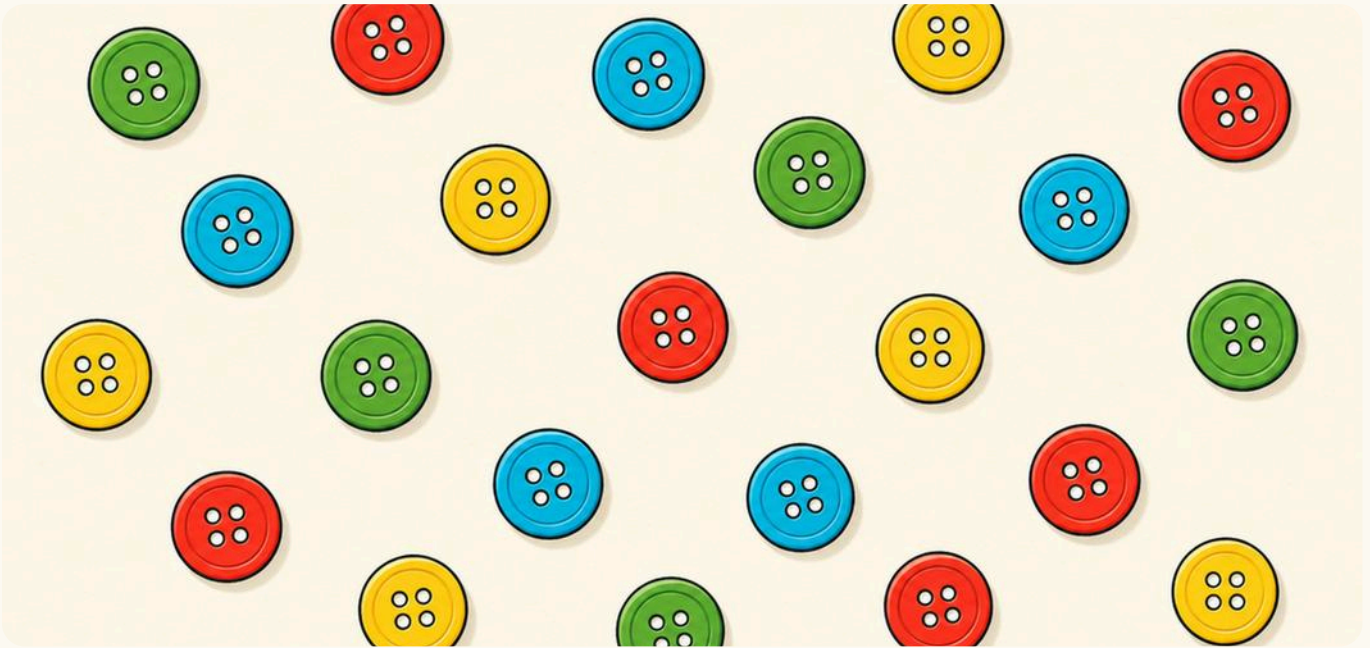
By the end you will

- ✓ say what data is
- ✓ collect data from your own class
- ✓ sort things into groups
- ✓ draw a chart and read what it tells you



Here is how it works

Data means facts you have collected. Not what you think. Not what you hope. What you actually found out.



A pile of buttons is data waiting to be sorted.

- 1 Look at the buttons. Say one thing you can tell just by looking. Now say one thing you would have to **count** to know.

NO SCREEN NEEDED



Chip's fact card

Everything a computer holds is data: your name, the score in a game, a photograph, a song. Underneath, all of it is stored as nothing but numbers.

Asking, and writing it down



Rui asks the question. Tomás answers. Somebody must write it down, or it is lost.



Tally marks

You cannot write a whole word every time somebody answers, so you draw one small line instead: a **tally mark**. Every fifth mark goes **across** the other four, which makes them easy to count in fives.

four

||||

five

|||||

3.2 • COLLECTING DATA • CONTINUED

- 1 Draw seven tally marks. Remember to put the fifth one across.
NO SCREEN NEEDED

seven tally marks

- 2 Now count them back in fives and ones. Five, six, seven.

Chip's fact card

Tally marks are one of the oldest ways of counting in the world. Marks cut into bone in gates of five have been found that are thousands of years old, long before anybody wrote numbers down.

Our favourite fruit

Class 1 asked everybody one question: **which fruit do you like best?**
Here is what they found.

Fruit	Tally	How many
 apple		7
 banana		5
 orange		4
 grapes		2
Altogether		18

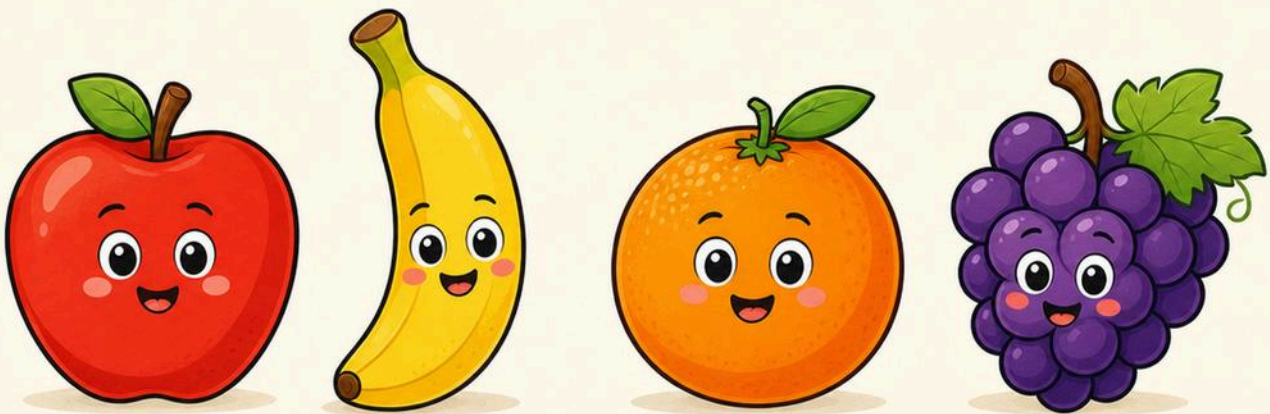
3.2 • YOUR TURN • CONTINUED

- 1 Count the tally marks in each row. Do they match the number beside them? Check every row.

NO SCREEN NEEDED

- 2 How many children answered altogether? Look at the bottom row.

18 children.



The four fruits Class 1 chose between.

Now ask your own class

Ask everybody in your class the same question. Put one tally mark in the row each time somebody answers.

Fruit	Tally	How many
 apple		
 banana		
 orange		
 grapes		
Altogether		

3.2 • YOUR TURN • CONTINUED

- 1 Ask every single person, and ask them all the **same** question in the **same** words. If you change the question halfway through, your data is spoilt.

NO SCREEN NEEDED

- 2 Count your marks and write the number in the last column. Then add them all up for the bottom row.

Robot challenge

Your bottom number should be the same as the number of children in your class. If it is not, somebody was asked twice, or not at all. Can you find out which?

For teachers

The check in the challenge above is the real lesson of this page: data has to add up. Ask pupils to compare their total against the class register before they go any further.

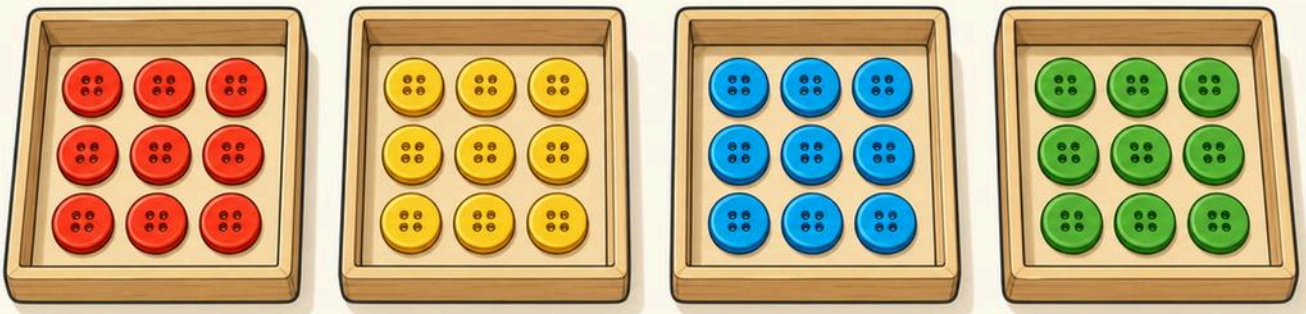
Putting like with like



In the yard today

Inês tipped the button box out onto the table.

There were buttons everywhere. So she did the thing that always helps: she put all the ones that were **the same** together.



Four trays, four colours. Now you can count them.



Here is how it works

To **sort** is to put things in groups, so that everything in one group is the same in some way. You choose **what** to sort by: colour, size, shape, or anything else you can see.

3.3 • SORTING AND GROUPING • CONTINUED

- 1 Sort the pencils on your table by colour. Now sort the same pencils again by **length**. Same pencils, different groups.

NO SCREEN NEEDED

- 2 Which is easier to count: a pile, or four tidy trays? Say why.



Chip's fact card

Computers sort enormous piles of data all day long. When you look for something on a computer, it can find it fast mostly because everything was sorted first.

Sort them yourself

Here are the buttons Inês counted after she had sorted them.

Colour	Tally	How many
red		6
yellow		4
blue		5
green		3
Altogether		18

1

Which colour had the most? Which had the fewest?

NO SCREEN NEEDED

3.3 • YOUR TURN • CONTINUED

2 Now sort something real. Tip out a box of bricks or buttons and put them in these four trays. Write what each tray is for at the top.

NO SCREEN NEEDED

[illegible]

Find the bug

Tomás sorted the buttons into **red**, **round** and **big**. Then he found a big round red button and did not know which tray to put it in.

3 Why did Tomás get stuck? What must be true about your groups?

Each thing must fit into exactly one group.

One picture for each one

A table of numbers is exact, but slow to read. A **pictogram** shows the same thing in pictures, and you can see the answer without counting at all.



 one child

3.4 • DISPLAYING DATA • CONTINUED

- 1 Which row is the **longest**? You did not have to count, did you? That is what a chart is for.

NO SCREEN NEEDED

- 2 Count the children in the apple row. Does it match the 7 in the table on the page before?



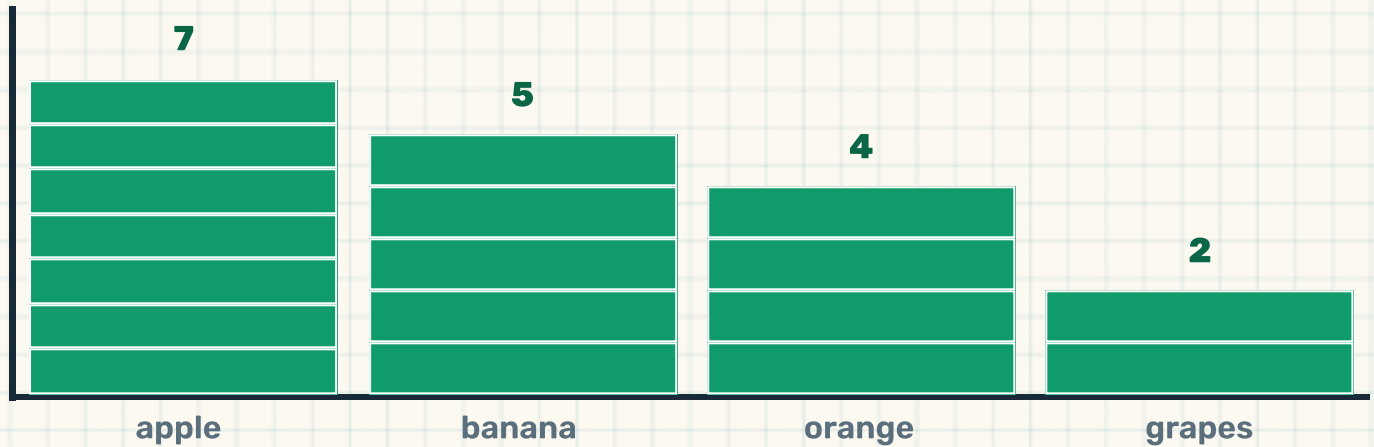
The key

The little note at the bottom is the **key**. It tells you what one picture means. Without a key, a chart is just a pattern.

Every chart needs a key, or nobody knows what they are looking at.

A block chart

Now the same data again, drawn as towers. One block means one child, exactly as before.



1 Put your finger on the **tallest** tower. Which fruit is it? NO SCREEN NEEDED

2 Put your finger on the **shortest**. Which fruit is that?

3 Count the blocks in the banana tower. It should be 5. Charts and tables must always agree.

→ One step more

Why do all the blocks have to be the same size? Draw one tower with big blocks and one with small blocks and see what goes wrong.

Draw your own chart



Use the data **you** collected from your class. Colour in one square for each child, starting at the bottom.

[illegible]

- 1 Colour from the **bottom** up, never from the middle. A tower with a gap in it cannot be counted.

NO SCREEN NEEDED

- 2** Write a title at the top of your chart, and remember your key.

my title and key

What is the chart telling you?

Reading a chart means answering questions with it. Look back at the fruit chart, then answer these.

More children chose apple than grapes.

TRUE / NOT TRUE

Orange was the most popular fruit.

TRUE / NOT TRUE

Exactly 5 children chose banana.

TRUE / NOT TRUE

Nobody chose grapes at all.

TRUE / NOT TRUE

- How many more children chose apple than grapes? Count the difference between the two towers.

5 more.

NO SCREEN NEEDED

2 How many chose apple **or** orange? Add the two numbers together.

11 children.

NO SCREEN NEEDED



Careful

This chart tells you what **Class 1** likes. It does not tell you what the whole school likes, and it certainly does not tell you which fruit is best. A chart only ever answers the question you actually asked.



Chip's fact card

This is the most important idea on the page, and adults get it wrong all the time: data can only tell you about the people you asked.

Ask your chart a question



Five kinds of weather. A whole term of weather is a lot of data.



A weather chart for your class

Put one mark on a chart every morning for a whole month, for the weather outside your window. At the end you will know something true about your own month that nobody else knows.

1

Write two questions your weather chart will be able to answer.

NO SCREEN NEEDED

3.5 • YOUR TURN • CONTINUED

my two questions

2 Now write one question it will **not** be able to answer.

Anything about next month, or about anywhere else.

NO SCREEN NEEDED

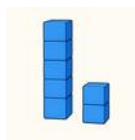
⚙ CHART WORDS



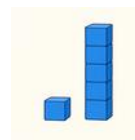
chart



pictogram



most



fewest

The words from Unit 3

Point at each picture and say the word. These are the words of the table.



data



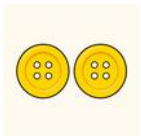
collect



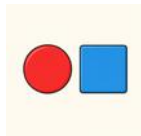
sort



group



same



different



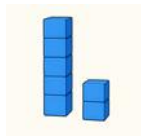
tally



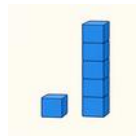
chart



pictogram



most



fewest



total



label



question

- 1 Say the difference between **most** and **fewest** without using either word.

NO SCREEN NEEDED

Our class survey poster



Your class is going to find out something true about itself, and put it on the wall for everybody to read.

- 1** Choose ONE question everybody agrees on. Favourite fruit, how you come to school, or how many letters are in your name.
- 2** Ask every single person the same question in the same words, and keep a tally as you go.
- 3** Count your tallies. Check that the total matches the number of children in your class.
- 4** Draw it as a big block chart. Give it a title and a key, then read it out to the class and say ONE true thing you learnt.

If your total does not match your class, do not fix the chart. Go back and find out who was missed.

Think about this unit

Colour one circle for each line. Green means yes, yellow means nearly, red means not yet. Not yet is a perfectly good answer: it tells you what to practise next.

I can say what data is.	☐	☐	☐
I can keep a tally while I ask people.	☐	☐	☐
I can sort things into groups that do not overlap.	☐	☐	☐
I can draw a chart with a title and a key.	☐	☐	☐
I can answer a question using a chart.	☐	☐	☐

Now I can

- ☐ say what data is, and what it is not
- ☐ collect data with tally marks
- ☐ sort and group things
- ☐ draw a pictogram and a block chart
- ☐ read a chart and say what is true

Next, in the Wire Garden, you will send your data to somebody else.

4

UNIT 4

The Wire Garden

Networks and digital communication

In the garden the cables grow along the ground like vines, joining everything to everything else. That is how a message gets from you to somebody far away.

1 What is a network

2 What is digital communication

3 Devices that connect

4 How we use networks

5 Staying safe

What is a network?



Before we begin

Hold hands in a ring with three friends. Now squeeze the hand of the person on your left, and ask them to pass the squeeze on. How far can one squeeze travel?



By the end you will

- ✓ say what a network is
- ✓ name devices that join a network
- ✓ say how people use networks every day
- ✓ know what to do if something worries you



Here is how it works

A network is two or more machines joined together so they can share. They might share a message, a picture, a game, or a printer. The joining is the important part.

The smallest network there is



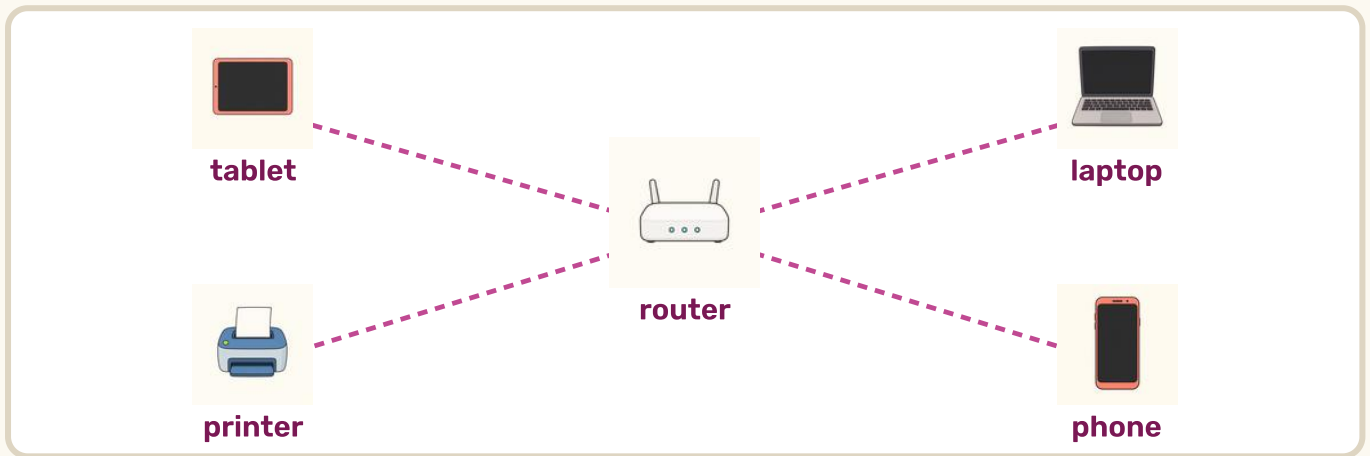
- 1 Two machines and one join. Trace the join with your finger. What could the tablet send to the printer?

Chip's fact card

The word **network** is much older than computers. A fishing net is a network: knots joined by threads. Change the knots into machines and the threads into cables, and you have the internet.

The box in the middle

When lots of machines join up, they do not all wire themselves to each other. They all join to one box in the middle. That box is called a **router**.



4.1 • HERE IS HOW IT WORKS • CONTINUED

- 1 Count the machines joined to the router. Now count the joins. Are they the same number?

NO SCREEN NEEDED

- 2 If the phone sends a picture to the printer, whose box does it go through on the way?

The router in the middle.

→ One step more

Imagine joining every machine straight to every other machine instead. With 5 machines that is 10 cables. With 20 machines it is 190. Now you can see why the box in the middle is such a good idea.

Sending a message a long way



In the yard today

Ada waved at her grandmother, who lives four hours away.

And her grandmother waved back, straight away. The wave went out of Ada's tablet, along a cable, and into her grandmother's room. That is **digital communication**.



Ada in Lisbon. Her grandmother far away. The little dots are the message travelling.



Here is how it works

To **communicate** is to send something to somebody and have them understand it. When machines do it, we call it digital. You can send words, a voice, a picture, or your whole moving face.



a message



a photo



a video call



sharing

Fast, and not so fast



Put these four ways of sending a message in order, from the **slowest** to the **fastest**. Write 1, 2, 3, 4.

☐

Walk to your friend's house and tell them.

☐

Put a letter in the post.

☐

Send a message from a tablet.

☐

Shout across the classroom.

4.2 • YOUR TURN • CONTINUED

- 1 Two of these are digital and two are not. Which two are digital? How do you know?

The ones that need a machine.

NO SCREEN NEEDED

Chip's fact card

A message from Lisbon to Australia goes most of the way along cables lying on the floor of the sea, and it gets there in less time than it takes you to say the word Australia.

- 2 Write a short message you would like to send to somebody far away. Draw a picture to go with it.

my message

What can join a network?



Six things that can join a network. Some you already have at school.



Two ways to join

A machine can join with a **cable**, which you can see and trip over, or with **wifi**, which you cannot see at all. Wifi is radio: the message travels through the air, like sound, but far too high for your ears.



a cable

you can see the join



wifi

through the air

4.3 • DEVICES THAT CONNECT • CONTINUED

- 1 Look round your classroom and find one thing joined by a cable and one joined by wifi. Point at each.

NO SCREEN NEEDED

- 2 Circle YES if the thing could join a network.

A tablet.

YES / NO

A wooden chair.

YES / NO

A printer.

YES / NO

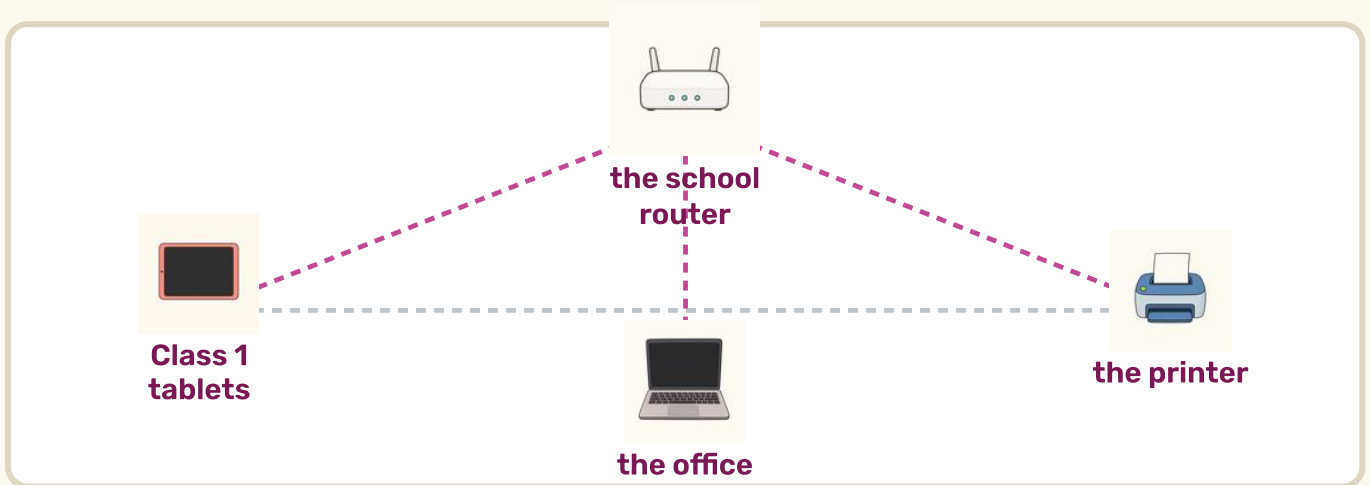
A pencil sharpener.

YES / NO

A network in your school



Your school has a network. Here is what it might look like. The faint lines are joins that are switched off right now.



- 1 Follow the line from Class 1 to the printer. Does it go straight there, or through the router?

NO SCREEN NEEDED

4.4 • HOW WE USE NETWORKS • CONTINUED



What a school network is for

1. Everybody can use **one** printer instead of twenty.
2. Your work is kept in one safe place, not on one tablet.
3. Your teacher can send the same thing to the whole class at once.

2

Name one thing your class shares that is NOT a machine. A ball? Books? Sharing is not only for computers.

NO SCREEN NEEDED

→ One step more

What happens to all of this if the router is switched off? Talk about which things would still work and which would not.

Draw your own network



Draw a box in the middle for the router. Then draw three machines round it and join each one to the router with a line.

my network

- 1 Label each machine you drew. A drawing of a network with no labels is just a spider.

NO SCREEN NEEDED

4.4 • YOUR TURN • CONTINUED



Find the bug

Rui drew four machines and a router, but he joined the machines to **each other** in a ring and forgot to join any of them to the router at all.

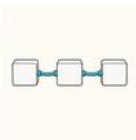
2

What is wrong with Rui's drawing? What must he add?

A line from every machine to the router.

NO SCREEN NEEDED

⚙️ NETWORK WORDS



network



router



cable



wifi

Being safe, and being kind



In the yard today

A network joins you to people. That is the whole point of it.

Most of those people are lovely. A few are not. So there are four rules in the Wire Garden, and they are easy.

Stop and think



If anything on a screen makes you feel worried, unhappy or muddled, tell a grown-up you trust straight away. You will never, ever be in trouble for telling.



The four rules of the garden

1. **Ask first.** Ask a grown-up before you go online.
2. **Keep it private.** Your name, your school and where you live are yours. Do not type them in.
3. **Be kind.** Only send a message you would be happy to say out loud, in front of everybody.
4. **Tell somebody.** If something feels wrong, tell a grown-up you trust.



Inês told her teacher. That was exactly the right thing to do.

Would you tell somebody?



Read each one with your teacher. Circle **YES** if you should tell a grown-up.

A message on the screen says something unkind about you.

YES / **NO**

You cannot work out how to close a game.

YES / **NO**

Somebody you do not know asks you where you live.

YES / **NO**

A picture comes up that you do not like.

YES / **NO**

1 Did you circle YES four times? Good. Telling somebody is always allowed, even for a small thing.

NO SCREEN NEEDED

4.5 • YOUR TURN • CONTINUED



Being kind on a screen is exactly the same as being kind in the corridor.

- 2** Name three grown-ups you could tell. Write their names or draw them.

NO SCREEN NEEDED

three people I trust

For teachers

Keep the tone here calm and matter of fact. The single message that protects children this age is **telling is always allowed and never gets you into trouble**. Frightened pupils stop disclosing, which is the opposite of what this page is for.

Ask pupils to name their three trusted adults out loud, and check that at least one is somebody in school.

A secret worth keeping



A password is a key. You would not lend somebody your front door key.



What makes a good password

A good password is **easy for you** to remember and **hard for anybody else** to guess. Three silly words joined together work beautifully, because they are easy to picture and very hard to guess.

blue

tractor

sock

1

Make up three silly words of your own that go together. Say them out loud, then draw them.

NO SCREEN NEEDED

4.5 • YOUR TURN • CONTINUED

word 1	word 2	word 3
--------	--------	--------

Stop and think



Never tell anybody your password, not even your best friend. Your teacher and the grown-ups at home are the only exception.

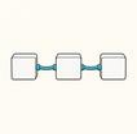


Chip's fact card

The most common password in the world is **123456**, which is the first thing anybody would try. Three silly words are millions of times harder to guess and much easier to remember.

The words from Unit 4

Point at each picture and say the word. These are the words of the garden.



network



cable



router



wifi



tablet



laptop



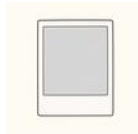
phone



printer



message



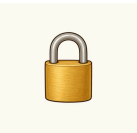
photo



video



share



password



safe



kind



ask

1 Choose **safe** and **kind**. Say one sentence that uses both.

NO SCREEN NEEDED

Build the class network



Your class is going to make a network out of paper and wool, on the floor, big enough to walk round.

- 1** Draw one machine each on a piece of paper: a tablet, a laptop, a printer, a phone. Cut them out.
- 2** Put one big paper box in the middle for the router, and lay your machines in a ring round it.
- 3** Join every machine to the router with a length of wool. Every machine needs exactly one piece.
- 4** Now pass a paper message from one machine to another. It must travel along the wool, through the router. No shortcuts.

→ One step more

Take one piece of wool away. Which machines can still send a message, and which cannot? That is what happens when a cable comes loose.

Think about this unit



Colour one circle for each line. Green means yes, yellow means nearly, red means not yet. Not yet is a perfectly good answer: it tells you what to practise next.

I can say what a network is.	☐ ☐ ☐
I can point at the router in a drawing.	☐ ☐ ☐
I can name three devices that connect.	☐ ☐ ☐
I know the four rules of the garden.	☐ ☐ ☐
I can name three grown-ups I could tell.	☐ ☐ ☐

Now I can

- ☐ say what a network is and what it is for
- ☐ name devices that join a network
- ☐ say how a message travels to somebody far away
- ☐ keep my private things private
- ☐ tell a grown-up I trust if something worries me

Next, in the Machine Room, you will look inside the machine itself.

5

UNIT 5

The Machine Room

Computer systems

In the machine room you find out what a computer is actually made of: the parts you can touch, the parts you cannot, and how they work together.

- 1 What is a computer system
- 2 Hardware: the parts you can touch
- 3 Software: the parts inside
- 4 Working together, and the rules

What is a computer system?



Before we begin

Name three computers in your school. Now think again: is there a computer inside the microwave in the staff room? Inside a car? Inside a watch?



By the end you will

- ✓ say what a computer system is
- ✓ name the parts you can touch
- ✓ say what software is
- ✓ know the rules for looking after a machine



Here is how it works

A **computer system** is all the parts working together to do a job. Some parts you can touch. Some parts you cannot. You need **both**, or nothing happens at all.

5.1 • SWITCH ON • CONTINUED



hardware

The parts you can **touch**. A screen, a keyboard, a mouse.



software

The parts you **cannot** touch. The instructions inside.

1

A guitar and the tune you play on it. Which one is like hardware? Which is like software?

NO SCREEN NEEDED



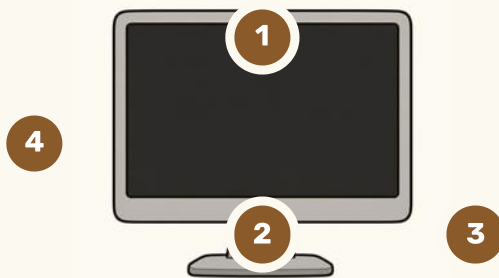
Chip's fact card

There is a computer inside a washing machine, a traffic light and a musical birthday card. Most of the computers in the world do not look anything like a computer.

The parts you can touch



Point at each numbered part on the picture, then find it in the list.



1. **The screen.** It shows you what the computer is doing.
2. **The keyboard.** You press the keys to put letters and numbers in.
3. **The mouse.** You move it to point at things, then click.
4. **The box.** The thinking part lives in here, out of the way.

1 Find all four parts on a real computer in your classroom. Touch each one gently as you name it.

2 Which of the four does the actual **thinking**?

The box. The screen only shows you the answer.

NO SCREEN NEEDED

You can touch every single one of these. That is what makes them hardware.

Things that go in, things that come out



Six pieces of hardware. Three send things IN, three send things OUT.

In, and out

An **input** puts something **into** the computer: your finger, your voice, a click. An **output** brings something **out**: a picture, a sound, a printed page. Your body works the same way. Ears in, voice out.

1 Sort these into the right tray. Write each one in.

NO SCREEN NEEDED

keyboard

screen

mouse

speaker

microphone

printer

5.2 • HARDWARE • CONTINUED

goes IN

.....

.....

.....

comes OUT

.....

.....

.....

- 2** Your eyes, your ears, your mouth, your fingers. Which are inputs and which are outputs?

NO SCREEN NEEDED

→ One step more

Here is a hard one. A touchscreen is both at once: it shows you a picture AND feels your finger. Can you think of another part that does two jobs?

The parts you cannot touch



In the yard today

Rui tried to find the drawing program inside the computer.

He looked all round the back. He looked underneath. He could not find it anywhere, because there was nothing to find. Software is not a **thing**. It is a set of instructions.



Here is how it works

Software is a program: a list of instructions the computer follows. You already know how to write one. Every script you built at the Block Bench was software.



Six pieces of software. Each one is a different set of instructions.

1

Point at the one you would open to draw a picture. Now the one for listening to music.

NO SCREEN NEEDED



hardware

you can touch it



software

you cannot



system

both, together

Hardware or software?



Read each one. Circle **YES** if you could pick it up and hold it in your hands.

A mouse.	YES / NO
A drawing program.	YES / NO
A pair of headphones.	YES / NO
A song stored on a tablet.	YES / NO
A printer.	YES / NO
A game.	YES / NO

5.3 • YOUR TURN • CONTINUED

- 1 Everything you said NO to is **software**. Say all of them out loud together.

NO SCREEN NEEDED



Find the bug

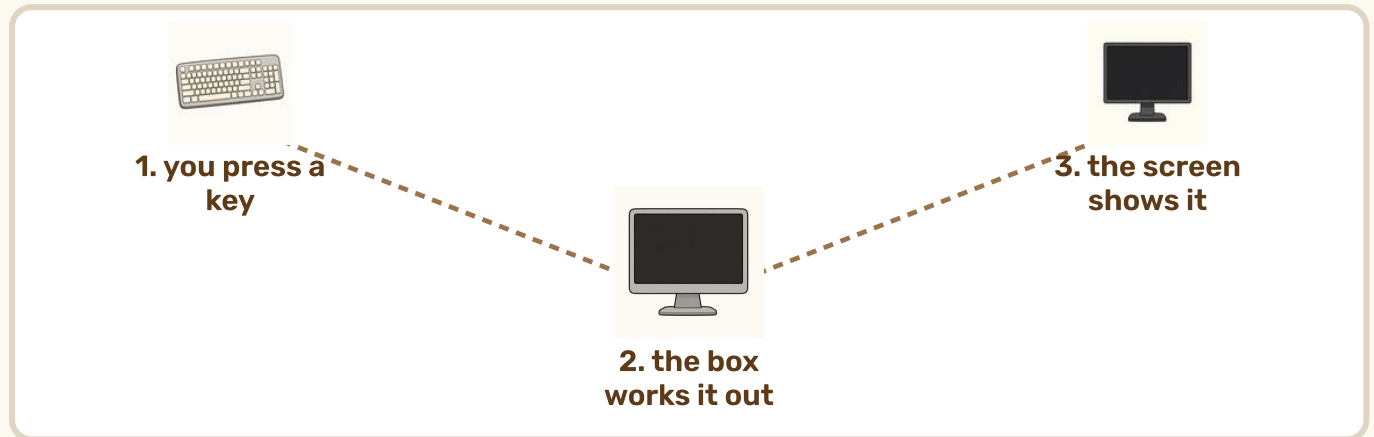
Ada's tablet would not play any music. She checked the speaker and it was fine. She checked the volume and it was up. Then she noticed she had never opened the music program at all.

- 2 Was Ada's problem in the hardware or the software?
The software. The hardware was working perfectly.

Perfect hardware with no software does nothing at all. It is a beautiful box that sits there.

From your finger to the screen

Here is what really happens when you press one key. It takes four steps, and it happens faster than you can blink.



- 1 Your finger presses a key. That is the **input** .
- 2 The key sends a message to the box.
- 3 The **software** in the box works out what to do.
- 4 The letter appears on the screen. That is the **output** .

5.4 • WORKING TOGETHER • CONTINUED

- 1 Press one key on a real keyboard and watch the screen. Try to spot the gap between pressing and appearing. You will not manage it.



In goes the key press. Out comes the letter on the screen.

Chip's fact card

Those four steps happen in about one thousandth of a second. In the time it takes you to blink, a computer could do it about four hundred times.

When one part is missing



A system only works when every part does its job. Say what would happen in each case.

What is missing	What happens
No screen	
No keyboard	
No software	
No electricity	

1

Talk about each row with your partner first, then draw or write your answer.

NO SCREEN NEEDED

5.4 • YOUR TURN • CONTINUED

2 Which one stops **everything**, all at once?

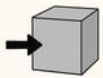
No electricity.

NO SCREEN NEEDED

⚙️ Robot challenge

Beep has wheels, a bumper, a battery and a set of instructions. Which of those four is Beep's software? Which are Beep's hardware?

⚙️ SYSTEM WORDS



input



output



system



charge

Looking after the machines



Two hands, flat underneath, walking slowly. That is how a tablet travels.

🛡️ Six rules of the machine room

1. **Clean dry hands.** Every time.
2. **No food, no drink** near a machine. Ever.
3. **Two hands, flat underneath.** Never carry a tablet by one corner.
4. **Press gently.** Pressing harder never helps.
5. **Ask before you plug anything in** or pull anything out.
6. **Finish properly.** Close what you opened, then put it back where it lives.

- 1 Say all six rules out loud together as a class. Then say the one you think is the most important, and why.

NO SCREEN NEEDED

A machine that is looked after works for years. A dropped one works for nobody.

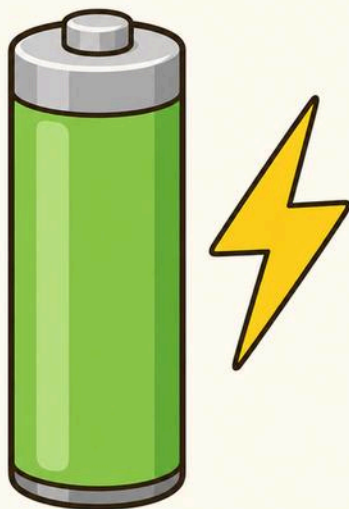
Good idea, or bad idea?



Circle **YES** if it is a good idea in the machine room.

Carrying a tablet flat, with both hands.	YES / NO
Having a drink of juice beside the keyboard.	YES / NO
Asking before you unplug a cable.	YES / NO
Pressing the screen really hard when it is slow.	YES / NO
Putting the tablet back on its shelf when you finish.	YES / NO
1 For every NO, say what you should do instead. NO SCREEN NEEDED	

5.5 • YOUR TURN • CONTINUED



A tablet with a flat battery is not broken. It is hungry.

2 Draw yourself carrying a tablet the right way.

carrying it properly

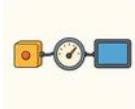
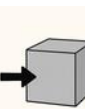
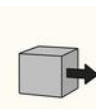
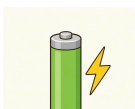
For teachers

Turn these six rules into a visible routine rather than a list on a wall: hands checked at the door, two-hand carry modelled by an adult, and a named pupil responsible for the trolley each week.

Pupils who have broken something are often too frightened to say. Make reporting damage completely blame-free, exactly as with the online-safety rule in Unit 4.

The words from Unit 5

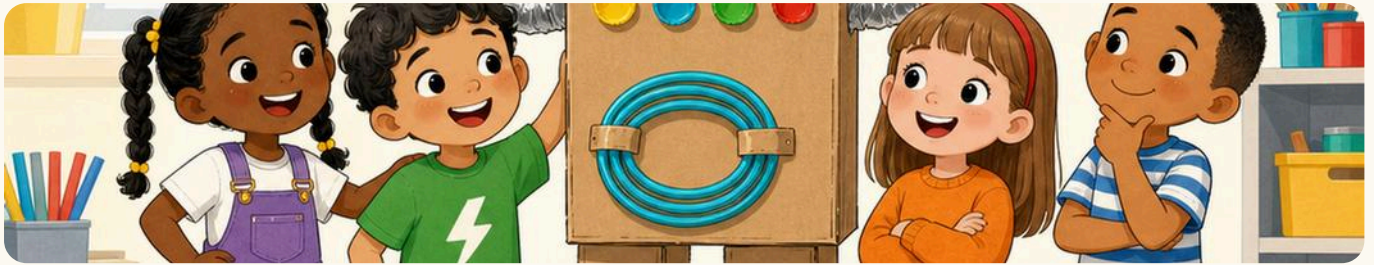
Point at each picture and say the word. These are the words of the machine room.

**computer****system****hardware****software****keyboard****mouse****screen****speaker****headphones****microphone****input****output****button****charge****care****printer**

- 1 Say the difference between hardware and software, out loud, in one sentence.

NO SCREEN NEEDED

Build a cardboard computer



Your class is going to build a computer system out of boxes, big enough to stand beside.

- 1** Decide together which parts you need. You must have at least one input and one output.
- 2** Make each part out of boxes, foil and tape. Give every part a label so everybody knows what it is.
- 3** Join your parts up with cable or wool, so you can see which part talks to which.
- 4** Now show it working. One person is the input, one is the box that thinks, one is the output. Pass a paper message all the way through.

→ One step more

Where is the software in your cardboard computer? It is the rules you agreed for passing the message. You cannot touch it, and nothing works without it.

Think about this unit

Colour one circle for each line. Green means yes, yellow means nearly, red means not yet. Not yet is a perfectly good answer: it tells you what to practise next.

I can say what a computer system is.	☐ ☐ ☐
I can name four pieces of hardware.	☐ ☐ ☐
I can say what software is.	☐ ☐ ☐
I can say which parts are inputs and which are outputs.	☐ ☐ ☐
I know the six rules of the machine room.	☐ ☐ ☐

Now I can

☐ say what a computer system is	
☐ name the parts I can touch	
☐ explain what software is	
☐ sort parts into inputs and outputs	
☐ look after a machine properly	

That is all five places. Now the whole yard is yours.

THE WHOLE YEAR

Open day in the yard



Everything you have learnt this year, shown to somebody who has never seen it. Four stations, one class, one afternoon.

Four stations, one class

Split into four teams. Each team takes one place in the yard and shows a visitor what happens there.

1 The Instruction Shed. Lay a grid on the floor and invite your visitor to programme Beep with cards. Put ONE bug in on purpose and let them find it.

2 The Block Bench. Show a script running on a screen. Explain what each colour of block does before you press the green flag.

3 The Sorting Table. Ask your visitor your survey question, add their answer to your tally, and show them how the chart changes.

4 The Machine Room. Show your cardboard computer. Ask your visitor to point at an input and an output.

Practise once before the visitors come. Once is enough to make the second time feel easy.

→ One step more

Give every visitor one small card as they leave, with one true thing your team taught them written on it.

Plan your station



My team is at	
The place in the yard we chose	
The one thing we will show	
What our visitor will DO	
My job in the team	

- 1 Fill this in with your team before you build anything. Agree it out loud first.

NO SCREEN NEEDED

- 2 Draw your station, with your visitor in the picture.

our station

Robot challenge

Can your whole team explain your station without any of you reading from this book? Practise until you can.

My computing portfolio

This is your own record of the year. Tick a box each time you finish one, and write the date beside it.



Everything you make this year belongs in the yard. This is where it all ends up.

ASSESSMENTS • CONTINUED

What I made	Where	Date
A route for Beep, with a bug I fixed	Unit 1	
A script that runs from a green flag	Unit 2	
A survey, a tally and a chart	Unit 3	
A network drawing with a router	Unit 4	
A cardboard computer system	Unit 5	
My open day station	Open day	

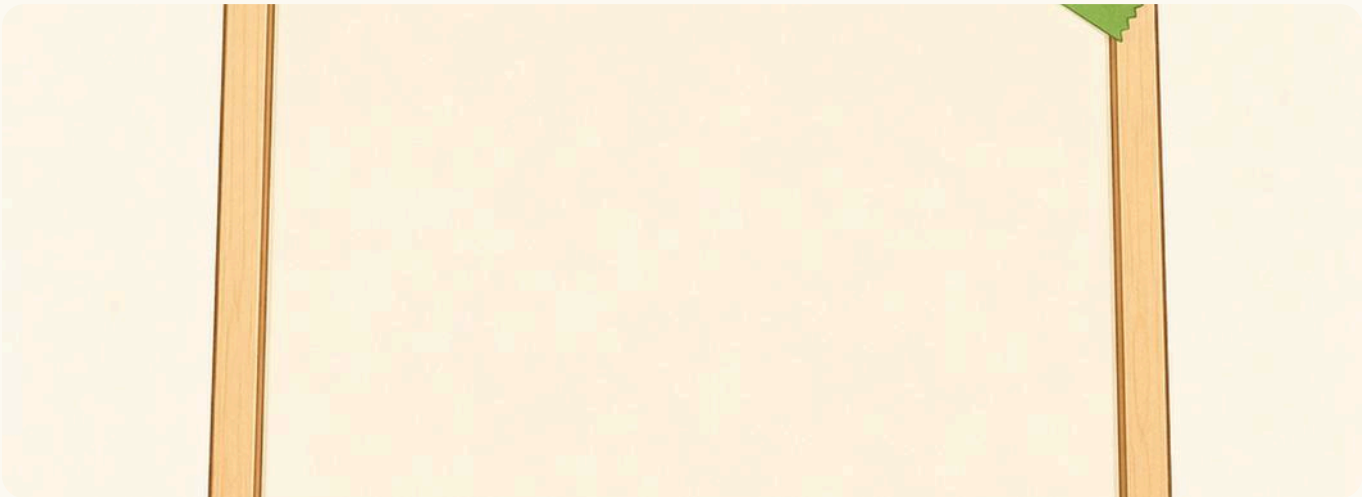
- 1 Look back through the whole book. Which one of these six were you most proud of?

NO SCREEN NEEDED

For teachers

This page is a pupil-facing record, not a mark sheet. Six pieces of evidence across the year is plenty at this age, and each maps to one unit of the scheme of work.

My best work this year



Choose it yourself. Nobody else can decide what you are most proud of.

draw or stick in my best work

What I made	
The hardest part	
What went wrong, and how I fixed it	

The third row is the most important one on this page. Everybody's work goes wrong. Fixing it is the skill.

How did the whole year go?

Colour one circle for each line. Green means yes, yellow means nearly, red means not yet.

I can put steps in the right order.	☐	☐	☐
I can find a bug and fix one step.	☐	☐	☐
I can build a script out of blocks.	☐	☐	☐
I can collect data and draw a chart.	☐	☐	☐
I can say what a network is.	☐	☐	☐
I can tell hardware from software.	☐	☐	☐
I know how to look after a machine.	☐	☐	☐
I know who to tell if something worries me.	☐	☐	☐

Next year I would like to

write or draw it

- 1 Look at your red circles. Those are not failures: they are your list for next year.

NO SCREEN NEEDED

A year in the yard



Beep works outside all year round, in every kind of weather.

1 Autumn term • Robot Sports Day: race Beep along a chalked grid

2 Spring term • Survey Week: every class asks one question and charts it

3 Summer term • Yard Open Day: show your station to visitors



Robot Sports Day

Two teams, one grid, one prize. Each team writes an algorithm in cards to get their robot from one corner to the other. The team whose robot gets there with the **fewest cards** wins. Not the fastest: the fewest.

1 Why is fewest cards a better prize than fastest? Talk about it.
Because a shorter algorithm is a cleverer one.

NO SCREEN NEEDED

Things to do in any week



Bug of the week

Your teacher puts a four-step algorithm on the wall with one thing wrong. Who spots it first?



Be the robot

One person is the robot and does exactly what the cards say. Nothing else. No guessing.



Weather chart

One mark every morning, all term. By the end you know something true about your own year.



Sensor hunt

Find every sensor in the school. Taps, lights, doors, bells. Make a list and add to it.

1

Choose one of these four to start this week. Write which one.

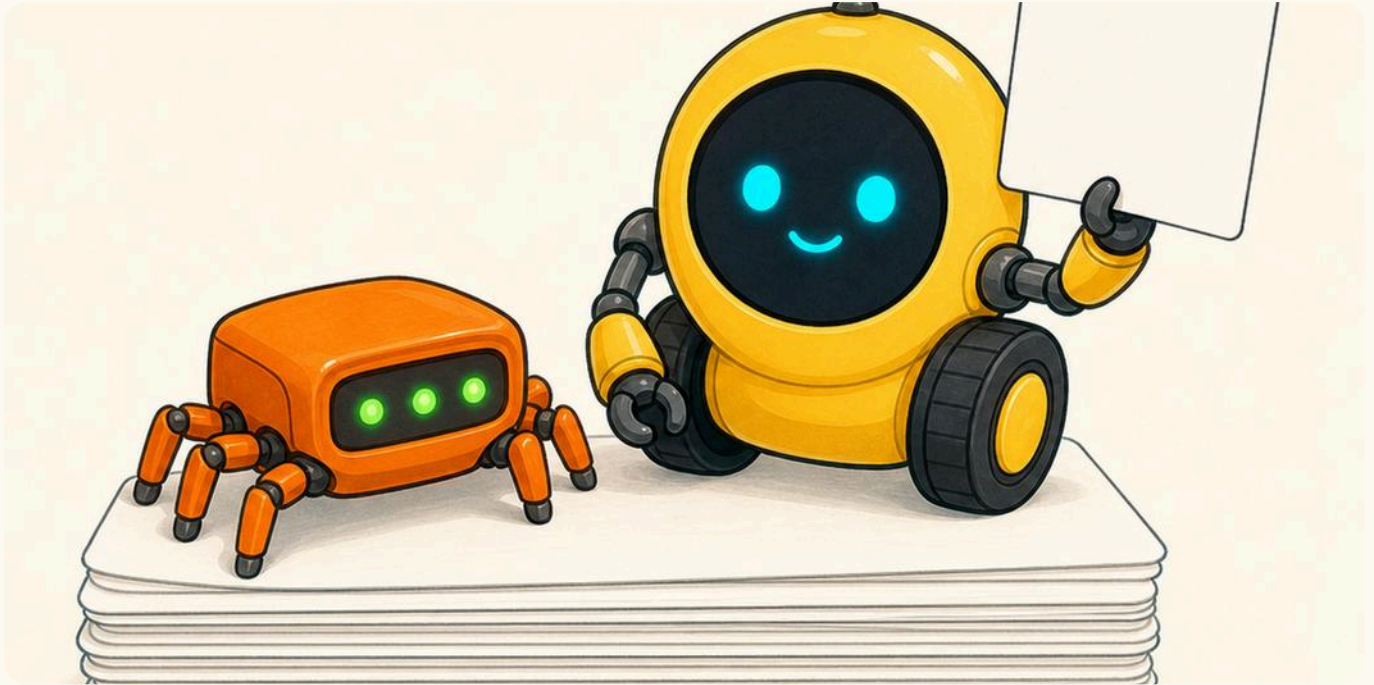
NO SCREEN NEEDED

this week we will

→ One step more

Keep a yard notebook for the whole class. Every time somebody finds a bug or a sensor, they write one line in it. By July it will be full.

Every word you learnt



Beep and Bolt kept every word you met this year.

There are **76** words in this book, and every single one has its own picture. They are in the order you met them, place by place.

How to use your word book

1. **Look at the picture first.** It tells you the word before you can read it.
2. **Say the word out loud.** Ears learn faster than eyes.
3. **Read the line underneath.** Your teacher will help.

- 1 Find a word you had forgotten. Say it out loud three times.

NO SCREEN NEEDED

The Instruction Shed



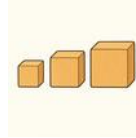
algorithm

a list of steps, in order,
that gets a job done



step

one thing you do out of a
longer list



order

which one comes first,
and which comes next



instruction

one thing you tell
somebody or something
to do



robot

a machine that follows
instructions



grid

squares in rows, so you
can say exactly where
something is



error

a step that does not do
what you wanted



bug

another word for an
error in an algorithm



debug

to look for a bug and fix
it



fix

to change the step that
was wrong



plan

to think out all your
steps before you start



sequence

things in a set order, one
after another

The Block Bench



block

one instruction you can snap to another one



script

blocks snapped together, that run from the top down



sprite

something on the screen that you can tell what to do



stage

the part of the screen where the sprite lives



move

to go from one place to another



turn

to face a different way without going anywhere



costume

a different picture for the same sprite



say

to put words in a speech bubble on the screen



sound

anything you can hear



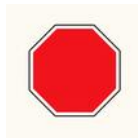
event

the thing that makes a script start



green flag

click it to start everything



stop

click it to stop everything

The Block Bench



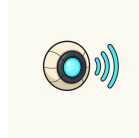
loop

blocks that happen again and again



forever

a loop that never stops on its own



sensor

the part of a machine that feels something



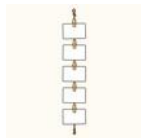
variable

a box with a name on it that holds one thing



backdrop

the picture behind everything on the stage



list

a box that holds many things, in order

The Sorting Table



data

facts you have collected,
not facts you guessed



collect

to gather data by asking
or counting



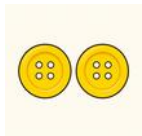
sort

to put things into groups



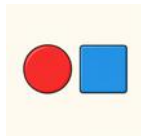
group

things that are the same
in some way



same

exactly alike in the way
you are looking at



different

not the same



tally

one small line for each
thing you count



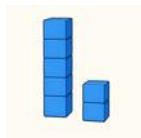
chart

a picture that shows you
what the numbers say



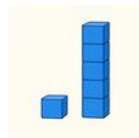
pictogram

a chart made of pictures,
one for each thing



most

the one there is more of
than any other



fewest

the one there is less of
than any other



total

how many there are
altogether

The Sorting Table



label

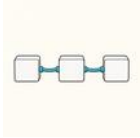
the words that say what something is



question

what you ask to find something out

The Wire Garden



network

machines joined together so they can share



cable

a wire that joins two machines



router

the box in the middle that everything joins to



wifi

a way of joining through the air, with no cable



tablet

a flat computer you hold in your hands



laptop

a computer that folds shut like a book



phone

a small computer you can also talk into



printer

it puts what is on the screen onto paper



message

words you send to somebody else



photo

a picture taken by a camera



video

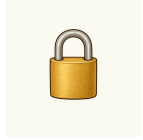
moving pictures, with sound



share

to let somebody else have it too

The Wire Garden



password

a secret only you know,
that unlocks your things



safe

nothing can hurt you
here



kind

thinking about somebody
else first



ask

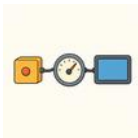
to say you need help, out
loud

The Machine Room



computer

a machine that follows a program



system

all the parts working together to do a job



hardware

the parts of a computer you can touch



software

the instructions inside, that you cannot touch



keyboard

you press its keys to put letters in



mouse

you move it to point, then you click



screen

it shows you what the computer is doing



speaker

the sound comes out of here



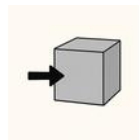
headphones

you wear them so only you can hear



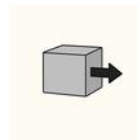
microphone

it hears your voice and puts it in



input

anything that goes into the computer



output

anything that comes out of the computer

The Machine Room



button

you press it to make something happen



charge

to fill a battery up again



care

to look after something properly



printer

it puts what is on the screen onto paper

Teaching Year 1 computing



The yard set out before a lesson. Most of it needs no devices.

For teachers

This book follows the Prime School scheme of work exactly. Five units, 31 topics, in the order the scheme sets them out. The unit titles in the running heads are the scheme's own, so you can map any page to the scheme in one glance. The yard names are for the pupils.

Unplugged first, every time. Around half the tasks in this book carry a green **No screen needed** flag. That is not a fallback for when the tablets are charging. Sequencing, spotting an error and fixing one step are learnt far better with cards on the floor than with a mouse, and pupils who have done it with their bodies transfer to the screen quickly.

Debugging is editing, not restarting. The commonest thing you will see is a pupil sweeping all their cards away and beginning again. Ask, every time: *which is the FIRST card that is wrong?* Then change only that one. This single habit is the most valuable thing in Unit 1.

For teachers • the two subject points

The block colours in this book are the real ones. They are taken from the Scratch editor itself, so a pupil who has met an orange Control block on paper will reach for the orange drawer on screen. Please do not recolour them in your own materials.

Charts must add up. In Unit 3, always have pupils check their survey total against the class register before they draw anything. Finding that the total is wrong, and then finding out why, is the lesson.

What you need, and what you do not

For teachers

You do not need a floor robot. Unit 1 works with a chalked grid and one child pretending to be Beep, and in some ways it works better: a child robot can be visibly, comically literal in a way a real one cannot. If you do have a floor robot, Unit 1 is the natural place for it.

You do not need one device each. Unit 2 works with one screen and a class gathered round it, with pupils taking turns to place a block. A two-minute rota with a visible timer works far better than asking pupils to judge when their turn is over.

What you do need: chalk or floor tiles for a grid; a stack of blank cards; coloured bricks or buttons for sorting; wool or string; cardboard boxes; and one screen with Scratch on it. All of it is ordinary classroom stock.

For teachers • tone

Beep is doing the teaching. Every joke in this book has the same shape: Beep does exactly what was said, not what was meant, and the result is funny. Lean into it. Play Beep yourself, in front of the class, and be gloriously literal. Pupils correct you eagerly, and correcting an instruction is precisely the skill being taught.

Two blame-free rules. In Unit 4, telling an adult about anything worrying on a screen must never carry consequences. In Unit 5, reporting damage to a device must never carry consequences either. Pupils who fear the consequence stop disclosing, which defeats the purpose of both pages.

Chip's fact card

If you only have time for one thing from this book, do **Bug of the week**: a four-step algorithm on the wall with one step wrong. Two minutes a week, and it teaches more than an hour of vocabulary.

PRIME SCHOOL PRESS

Computing & Robotics

Year 1 · Prime School Press · Student Manual

A robot does exactly what you say. Not what you meant.

Behind the school there is a yard, a garden and a machine room. Learn to say things precisely, build instructions out of blocks, and find out what a machine is really doing.

INSIDE THIS BOOK

- Every Year 1 computing topic, in five units
- Predict, run and compare on every unit
- Real bug hunts, with the fix never printed
- Projects along the way, and one big one at the end
- QR codes, each printed in words as well

Prime School Press · Computing & Robotics

Ages 5–6 · Lower Primary

primeschool.pt